

C5g7 comsol valid

Report date	Jun 29, 2026, 8:44:20 PM
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1 Global Definitions

Date Jun

GLOBAL SETTINGS

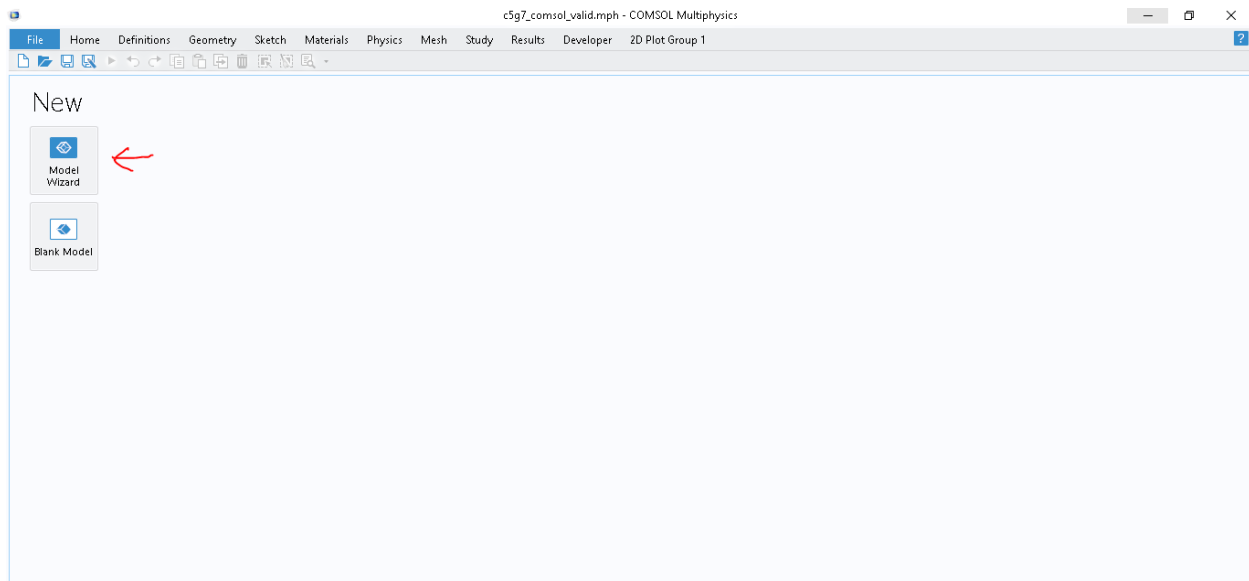
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Version	COMSOL Multiphysics 6.0
Unit system	SI

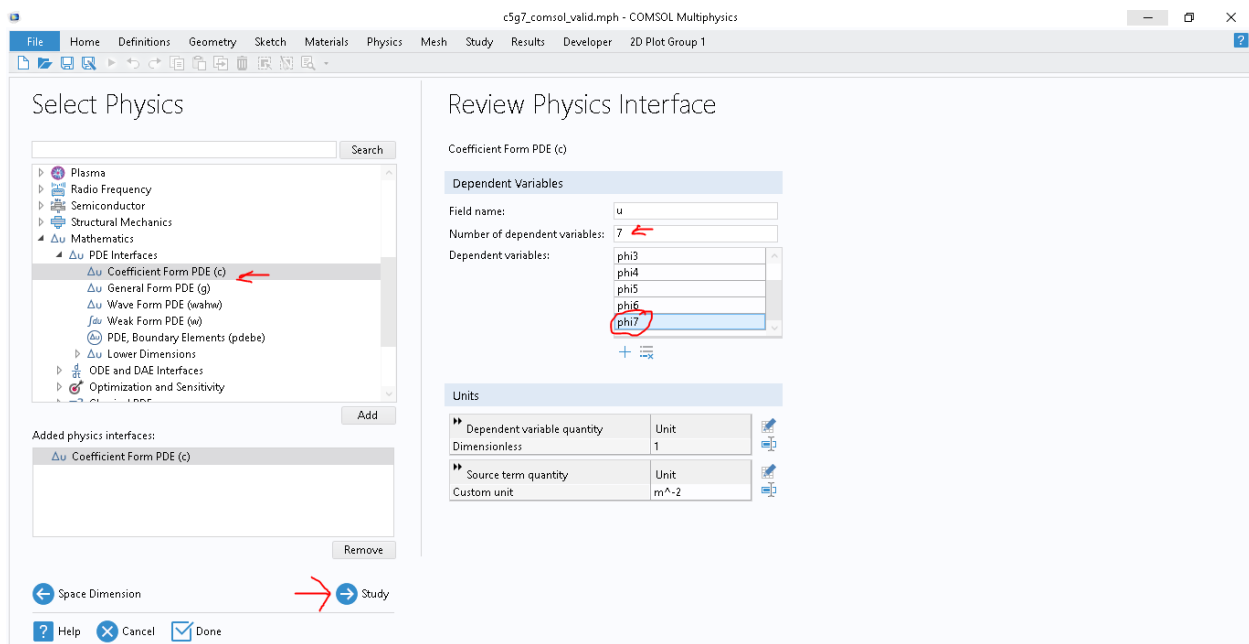
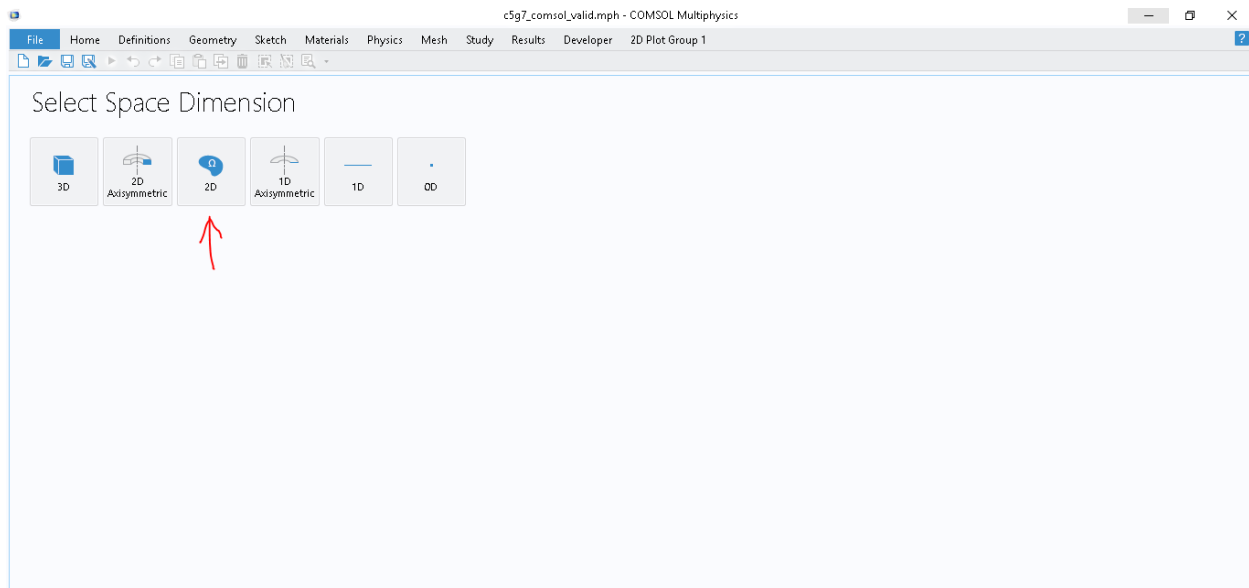
USED PRODUCTS

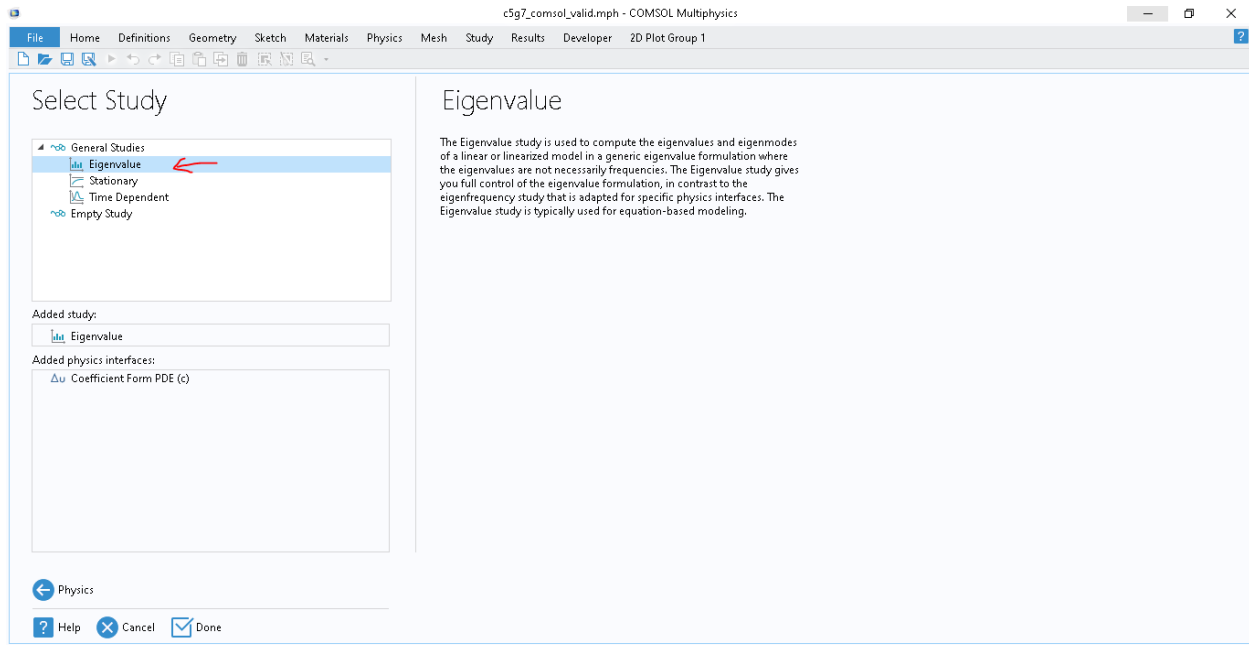
COMSOL Multiphysics

COMPUTER INFORMATION

CPU	Intel64 Family 6 Model 58 Stepping 9, 2 cores
Operating system	Windows 10







1.1 PARAMETERS

PARAMETERS 1

Name	Expression	Value	Description
Chi_g1	5.87910000E-01	0.58791	
Chi_g2	4.11760000E-01	0.41176	
Chi_g3	3.39060000E-04	3.3906E-4	
Chi_g4	1.17610000E-07	1.1761E-7	
Chi_g5	0.0	0	
Chi_g6	0.0	0	
Chi_g7	0.0	0	
D_Mod_g1	2.09372344	2.0937	
D_Mod_g2	0.80716113	0.80716	
D_Mod_g3	0.56467506	0.56468	
D_Mod_g4	0.57043439	0.57043	
D_Mod_g5	0.46425255	0.46425	
D_Mod_g6	0.26572070	0.26572	
D_Mod_g7	0.12576813	0.12577	
Srem_Mod_g1	1.14728300E-01	0.11473	
Srem_Mod_g2	1.30636000E-01	0.13064	
Srem_Mod_g3	2.45054000E-01	0.24505	

Name	Expression	Value	Description
Srem_Mod_g4	4.93321600E-01	0.49332	
Srem_Mod_g5	5.78862000E-01	0.57886	
Srem_Mod_g6	5.54537000E-01	0.55454	
Srem_Mod_g7	1.69680000E-01	0.16968	
NuSigF_Mod_g1	0.0	0	
NuSigF_Mod_g2	0.0	0	
NuSigF_Mod_g3	0.0	0	
NuSigF_Mod_g4	0.0	0	
NuSigF_Mod_g5	0.0	0	
NuSigF_Mod_g6	0.0	0	
NuSigF_Mod_g7	0.0	0	
Ss_Mod_g1_to_g2	1.134000E-01	0.1134	
Ss_Mod_g1_to_g3	7.234700E-04	7.2347E-4	
Ss_Mod_g2_to_g3	1.299400E-01	0.12994	
Ss_Mod_g1_to_g4	3.749900E-06	3.7499E-6	
Ss_Mod_g2_to_g4	6.234000E-04	6.234E-4	
Ss_Mod_g3_to_g4	2.245700E-01	0.22457	
Ss_Mod_g5_to_g4	7.143700E-05	7.1437E-5	
Ss_Mod_g1_to_g5	5.318400E-08	5.3184E-8	
Ss_Mod_g2_to_g5	4.800200E-05	4.8002E-5	
Ss_Mod_g3_to_g5	1.699900E-02	0.016999	
Ss_Mod_g4_to_g5	4.155100E-01	0.41551	
Ss_Mod_g6_to_g5	2.215700E-03	0.0022157	
Ss_Mod_g2_to_g6	7.448600E-06	7.4486E-6	
Ss_Mod_g3_to_g6	2.644300E-03	0.0026443	
Ss_Mod_g4_to_g6	6.373200E-02	0.063732	
Ss_Mod_g5_to_g6	5.118200E-01	0.51182	
Ss_Mod_g7_to_g6	1.324400E-01	0.13244	
Ss_Mod_g2_to_g7	1.045500E-06	1.0455E-6	
Ss_Mod_g3_to_g7	5.034400E-04	5.0344E-4	
Ss_Mod_g4_to_g7	1.213900E-02	0.012139	
Ss_Mod_g5_to_g7	6.122900E-02	0.061229	
Ss_Mod_g6_to_g7	5.373200E-01	0.53732	
D_UO2_g1	1.96053958	1.9605	
D_UO2_g2	0.91328822	0.91329	

Name	Expression	Value	Description
D_UO2_g3	0.63265271	0.63265	
D_UO2_g4	0.58783868	0.58784	
D_UO2_g5	0.68925761	0.68926	
D_UO2_g6	0.43939382	0.43939	
D_UO2_g7	0.23040685	0.23041	
Srem_UO2_g1	7.76160441E-02	0.077616	
Srem_UO2_g2	5.83419904E-02	0.058342	
Srem_UO2_g3	1.20643469E-01	0.12064	
Srem_UO2_g4	2.67404133E-01	0.2674	
Srem_UO2_g5	2.68154768E-01	0.26815	
Srem_UO2_g6	3.09201759E-01	0.3092	
Srem_UO2_g7	2.39873053E-01	0.23987	
NuSigF_UO2_g1	1.15751545E-02	0.011575	
NuSigF_UO2_g2	1.16980902E-03	0.0011698	
NuSigF_UO2_g3	9.06278378E-03	0.0090628	
NuSigF_UO2_g4	2.60718267E-02	0.026072	
NuSigF_UO2_g5	2.50095600E-02	0.02501	
NuSigF_UO2_g6	1.16611489E-01	0.11661	
NuSigF_UO2_g7	3.03349269E-01	0.30335	
Ss_UO2_g1_to_g2	7.241837E-02	0.072418	
Ss_UO2_g1_to_g3	3.114538E-04	3.1145E-4	
Ss_UO2_g2_to_g3	5.590244E-02	0.055902	
Ss_UO2_g1_to_g4	1.589289E-06	1.5893E-6	
Ss_UO2_g2_to_g4	2.636830E-04	2.6368E-4	
Ss_UO2_g3_to_g4	9.653297E-02	0.096533	
Ss_UO2_g5_to_g4	1.024886E-04	1.0249E-4	
Ss_UO2_g1_to_g5	2.249538E-08	2.2495E-8	
Ss_UO2_g2_to_g5	2.030354E-05	2.0304E-5	
Ss_UO2_g3_to_g5	7.190114E-03	0.0071901	
Ss_UO2_g4_to_g5	1.789614E-01	0.17896	
Ss_UO2_g6_to_g5	1.685470E-03	0.0016855	
Ss_UO2_g2_to_g6	3.150555E-06	3.1506E-6	
Ss_UO2_g3_to_g6	1.118467E-03	0.0011185	
Ss_UO2_g4_to_g6	2.695690E-02	0.026957	
Ss_UO2_g5_to_g6	2.224033E-01	0.2224	

Name	Expression	Value	Description
Ss_UO2_g7_to_g6	6.094967E-02	0.06095	
Ss_UO2_g2_to_g7	4.422180E-07	4.4222E-7	
Ss_UO2_g3_to_g7	2.129414E-04	2.1294E-4	
Ss_UO2_g4_to_g7	5.134467E-03	0.0051345	
Ss_UO2_g5_to_g7	2.589821E-02	0.025898	
Ss_UO2_g6_to_g7	2.369710E-01	0.23697	
D_MOX43_g1	1.95535009	1.9554	
D_MOX43_g2	0.91178329	0.91178	
D_MOX43_g3	0.63031671	0.63032	
D_MOX43_g4	0.58042324	0.58042	
D_MOX43_g5	0.60645862	0.60646	
D_MOX43_g6	0.36137692	0.36138	
D_MOX43_g7	0.22001288	0.22001	
Srem_MOX43_g1	7.72946399E-02	0.077295	
Srem_MOX43_g2	5.83696877E-02	0.05837	
Srem_MOX43_g3	1.21298972E-01	0.1213	
Srem_MOX43_g4	2.71989768E-01	0.27199	
Srem_MOX43_g5	3.31058237E-01	0.33106	
Srem_MOX43_g6	4.80387848E-01	0.48039	
Srem_MOX43_g7	3.12877961E-01	0.31288	
NuSigF_MOX43_g1	1.25520758E-02	0.012552	
NuSigF_MOX43_g2	1.46282361E-03	0.0014628	
NuSigF_MOX43_g3	9.38707394E-03	0.0093871	
NuSigF_MOX43_g4	3.77803379E-02	0.03778	
NuSigF_MOX43_g5	1.77286350E-02	0.017729	
NuSigF_MOX43_g6	3.84675752E-01	0.38468	
NuSigF_MOX43_g7	4.11991915E-01	0.41199	
Ss_MOX43_g1_to_g2	7.186154E-02	0.071862	
Ss_MOX43_g1_to_g3	3.107565E-04	3.1076E-4	
Ss_MOX43_g2_to_g3	5.590712E-02	0.055907	
Ss_MOX43_g1_to_g4	1.589014E-06	1.589E-6	
Ss_MOX43_g2_to_g4	2.636822E-04	2.6368E-4	
Ss_MOX43_g3_to_g4	9.649546E-02	0.096495	
Ss_MOX43_g5_to_g4	1.228057E-04	1.2281E-4	
Ss_MOX43_g1_to_g5	2.249538E-08	2.2495E-8	

Name	Expression	Value	Description
Ss_MOX43_g2_to_g5	2.030354E-05	2.0304E-5	
Ss_MOX43_g3_to_g5	7.190114E-03	0.0071901	
Ss_MOX43_g4_to_g5	1.789458E-01	0.17895	
Ss_MOX43_g6_to_g5	2.094178E-03	0.0020942	
Ss_MOX43_g2_to_g6	3.150555E-06	3.1506E-6	
Ss_MOX43_g3_to_g6	1.118467E-03	0.0011185	
Ss_MOX43_g4_to_g6	2.695690E-02	0.026957	
Ss_MOX43_g5_to_g6	2.218596E-01	0.22186	
Ss_MOX43_g7_to_g6	6.092025E-02	0.06092	
Ss_MOX43_g2_to_g7	4.422180E-07	4.4222E-7	
Ss_MOX43_g3_to_g7	2.129414E-04	2.1294E-4	
Ss_MOX43_g4_to_g7	5.134467E-03	0.0051345	
Ss_MOX43_g5_to_g7	2.589821E-02	0.025898	
Ss_MOX43_g6_to_g7	2.358406E-01	0.23584	
D_MOX70_g1	1.93834387	1.9383	
D_MOX70_g2	0.90674695	0.90675	
D_MOX70_g3	0.62350462	0.6235	
D_MOX70_g4	0.56659291	0.56659	
D_MOX70_g5	0.57738093	0.57738	
D_MOX70_g6	0.32950818	0.32951	
D_MOX70_g7	0.20657861	0.20658	
Srem_MOX70_g1	7.78780144E-02	0.077878	
Srem_MOX70_g2	5.86830135E-02	0.058683	
Srem_MOX70_g3	1.24086013E-01	0.12409	
Srem_MOX70_g4	2.82236618E-01	0.28224	
Srem_MOX70_g5	3.55565737E-01	0.35557	
Srem_MOX70_g6	5.71451399E-01	0.57145	
Srem_MOX70_g7	4.14566891E-01	0.41457	
NuSigF_MOX70_g1	1.37412986E-02	0.013741	
NuSigF_MOX70_g2	2.22656848E-03	0.0022266	
NuSigF_MOX70_g3	1.39259758E-02	0.013926	
NuSigF_MOX70_g4	5.44518778E-02	0.054452	
NuSigF_MOX70_g5	2.64104645E-02	0.02641	
NuSigF_MOX70_g6	5.35585937E-01	0.53559	
NuSigF_MOX70_g7	6.01954866E-01	0.60195	

Name	Expression	Value	Description
Ss_MOX70_g1_to_g2	7.208023E-02	0.07208	
Ss_MOX70_g1_to_g3	3.109189E-04	3.1092E-4	
Ss_MOX70_g2_to_g3	5.590948E-02	0.055909	
Ss_MOX70_g1_to_g4	1.589067E-06	1.5891E-6	
Ss_MOX70_g2_to_g4	2.636825E-04	2.6368E-4	
Ss_MOX70_g3_to_g4	9.644866E-02	0.096449	
Ss_MOX70_g5_to_g4	1.318823E-04	1.3188E-4	
Ss_MOX70_g1_to_g5	2.249538E-08	2.2495E-8	
Ss_MOX70_g2_to_g5	2.030354E-05	2.0304E-5	
Ss_MOX70_g3_to_g5	7.190114E-03	0.0071901	
Ss_MOX70_g4_to_g5	1.789096E-01	0.17891	
Ss_MOX70_g6_to_g5	2.250495E-03	0.0022505	
Ss_MOX70_g2_to_g6	3.150555E-06	3.1506E-6	
Ss_MOX70_g3_to_g6	1.118467E-03	0.0011185	
Ss_MOX70_g4_to_g6	2.695690E-02	0.026957	
Ss_MOX70_g5_to_g6	2.215227E-01	0.22152	
Ss_MOX70_g7_to_g6	6.113357E-02	0.061134	
Ss_MOX70_g2_to_g7	4.422180E-07	4.4222E-7	
Ss_MOX70_g3_to_g7	2.129414E-04	2.1294E-4	
Ss_MOX70_g4_to_g7	5.134467E-03	0.0051345	
Ss_MOX70_g5_to_g7	2.589821E-02	0.025898	
Ss_MOX70_g6_to_g7	2.348389E-01	0.23484	
D_MOX87_g1	1.92720835	1.9272	
D_MOX87_g2	0.90343291	0.90343	
D_MOX87_g3	0.61901349	0.61901	
D_MOX87_g4	0.55840057	0.5584	
D_MOX87_g5	0.56135889	0.56136	
D_MOX87_g6	0.31385652	0.31386	
D_MOX87_g7	0.19933428	0.19933	
Srem_MOX87_g1	7.82675078E-02	0.078268	
Srem_MOX87_g2	5.88918974E-02	0.058892	
Srem_MOX87_g3	1.25990780E-01	0.12599	
Srem_MOX87_g4	2.88379649E-01	0.28838	
Srem_MOX87_g5	3.70047966E-01	0.37005	
Srem_MOX87_g6	6.23132265E-01	0.62313	

Name	Expression	Value	Description
Srem_MOX87_g7	4.75191679E-01	0.47519	
NuSigF_MOX87_g1	1.45330100E-02	0.014533	
NuSigF_MOX87_g2	2.73482600E-03	0.0027348	
NuSigF_MOX87_g3	1.70096396E-02	0.01701	
NuSigF_MOX87_g4	6.47713048E-02	0.064771	
NuSigF_MOX87_g5	3.19113438E-02	0.031911	
NuSigF_MOX87_g6	6.20303593E-01	0.6203	
NuSigF_MOX87_g7	7.15108896E-01	0.71511	
Ss_MOX87_g1_to_g2	7.222680E-02	0.072227	
Ss_MOX87_g1_to_g3	3.110266E-04	3.1103E-4	
Ss_MOX87_g2_to_g3	5.591104E-02	0.055911	
Ss_MOX87_g1_to_g4	1.589103E-06	1.5891E-6	
Ss_MOX87_g2_to_g4	2.636827E-04	2.6368E-4	
Ss_MOX87_g3_to_g4	9.641508E-02	0.096415	
Ss_MOX87_g5_to_g4	1.375257E-04	1.3753E-4	
Ss_MOX87_g1_to_g5	2.249538E-08	2.2495E-8	
Ss_MOX87_g2_to_g5	2.030354E-05	2.0304E-5	
Ss_MOX87_g3_to_g5	7.190114E-03	0.0071901	
Ss_MOX87_g4_to_g5	1.788844E-01	0.17888	
Ss_MOX87_g6_to_g5	2.317199E-03	0.0023172	
Ss_MOX87_g2_to_g6	3.150555E-06	3.1506E-6	
Ss_MOX87_g3_to_g6	1.118467E-03	0.0011185	
Ss_MOX87_g4_to_g6	2.695690E-02	0.026957	
Ss_MOX87_g5_to_g6	2.213314E-01	0.22133	
Ss_MOX87_g7_to_g6	6.119335E-02	0.061193	
Ss_MOX87_g2_to_g7	4.422180E-07	4.4222E-7	
Ss_MOX87_g3_to_g7	2.129414E-04	2.1294E-4	
Ss_MOX87_g4_to_g7	5.134467E-03	0.0051345	
Ss_MOX87_g5_to_g7	2.589821E-02	0.025898	
Ss_MOX87_g6_to_g7	2.343819E-01	0.23438	
D_FC_g1	2.37986956	2.3799	
D_FC_g2	0.96945339	0.96945	
D_FC_g3	0.80572701	0.80573	
D_FC_g4	0.81435811	0.81436	
D_FC_g5	0.67111349	0.67111	

Name	Expression	Value	Description
D_FC_g6	0.38895922	0.38896	
D_FC_g7	0.18545355	0.18545	
Srem_FC_g1	8.30713171E-02	0.083071	
Srem_FC_g2	8.57126997E-02	0.085713	
Srem_FC_g3	1.61829941E-01	0.16183	
Srem_FC_g4	3.25188274E-01	0.32519	
Srem_FC_g5	3.80272888E-01	0.38027	
Srem_FC_g6	3.78156512E-01	0.37816	
Srem_FC_g7	1.13916089E-01	0.11392	
NuSigF_FC_g1	7.63637826E-09	7.6364E-9	
NuSigF_FC_g2	8.27745550E-09	8.2775E-9	
NuSigF_FC_g3	6.51232519E-07	6.5123E-7	
NuSigF_FC_g4	7.36459451E-06	7.3646E-6	
NuSigF_FC_g5	2.04181198E-07	2.0418E-7	
NuSigF_FC_g6	1.00408443E-06	1.0041E-6	
NuSigF_FC_g7	2.92166303E-06	2.9217E-6	
Ss_FC_g1_to_g2	8.205011E-02	0.08205	
Ss_FC_g1_to_g3	4.695030E-04	4.695E-4	
Ss_FC_g2_to_g3	8.521751E-02	0.085218	
Ss_FC_g1_to_g4	2.429835E-06	2.4298E-6	
Ss_FC_g2_to_g4	4.078803E-04	4.0788E-4	
Ss_FC_g3_to_g4	1.482397E-01	0.14824	
Ss_FC_g5_to_g4	5.176210E-05	5.1762E-5	
Ss_FC_g1_to_g5	3.440638E-08	3.4406E-8	
Ss_FC_g2_to_g5	3.140497E-05	3.1405E-5	
Ss_FC_g3_to_g5	1.119266E-02	0.011193	
Ss_FC_g4_to_g5	2.738383E-01	0.27384	
Ss_FC_g6_to_g5	1.466557E-03	0.0014666	
Ss_FC_g2_to_g6	4.874424E-06	4.8744E-6	
Ss_FC_g3_to_g6	1.741079E-03	0.0017411	
Ss_FC_g4_to_g6	4.187883E-02	0.041879	
Ss_FC_g5_to_g6	3.358094E-01	0.33581	
Ss_FC_g7_to_g6	8.475043E-02	0.08475	
Ss_FC_g2_to_g7	6.853773E-07	6.8538E-7	
Ss_FC_g3_to_g7	3.314801E-04	3.3148E-4	

Name	Expression	Value	Description
Ss_FC_g4_to_g7	7.976672E-03	0.0079767	
Ss_FC_g5_to_g7	4.002267E-02	0.040023	
Ss_FC_g6_to_g7	3.650428E-01	0.36504	
D_GT_g1	2.37986956	2.3799	
D_GT_g2	0.96945339	0.96945	
D_GT_g3	0.80573825	0.80574	
D_GT_g4	0.81442700	0.81443	
D_GT_g5	0.67112909	0.67113	
D_GT_g6	0.38895922	0.38896	
D_GT_g7	0.18545295	0.18545	
Srem_GT_g1	8.30713171E-02	0.083071	
Srem_GT_g2	8.57126997E-02	0.085713	
Srem_GT_g3	1.61898030E-01	0.1619	
Srem_GT_g4	3.25283945E-01	0.32528	
Srem_GT_g5	3.80272773E-01	0.38027	
Srem_GT_g6	3.78161705E-01	0.37816	
Srem_GT_g7	1.13910319E-01	0.11391	
NuSigF_GT_g1	0.0	0	
NuSigF_GT_g2	0.0	0	
NuSigF_GT_g3	0.0	0	
NuSigF_GT_g4	0.0	0	
NuSigF_GT_g5	0.0	0	
NuSigF_GT_g6	0.0	0	
NuSigF_GT_g7	0.0	0	
Ss_GT_g1_to_g2	8.205011E-02	0.08205	
Ss_GT_g1_to_g3	4.695030E-04	4.695E-4	
Ss_GT_g2_to_g3	8.521751E-02	0.085218	
Ss_GT_g1_to_g4	2.429835E-06	2.4298E-6	
Ss_GT_g2_to_g4	4.078803E-04	4.0788E-4	
Ss_GT_g3_to_g4	1.483026E-01	0.1483	
Ss_GT_g5_to_g4	5.175806E-05	5.1758E-5	
Ss_GT_g1_to_g5	3.440638E-08	3.4406E-8	
Ss_GT_g2_to_g5	3.140497E-05	3.1405E-5	
Ss_GT_g3_to_g5	1.119734E-02	0.011197	
Ss_GT_g4_to_g5	2.739248E-01	0.27392	

Name	Expression	Value	Description
Ss_GT_g6_to_g5	1.466465E-03	0.0014665	
Ss_GT_g2_to_g6	4.874424E-06	4.8744E-6	
Ss_GT_g3_to_g6	1.741829E-03	0.0017418	
Ss_GT_g4_to_g6	4.189094E-02	0.041891	
Ss_GT_g5_to_g6	3.358094E-01	0.33581	
Ss_GT_g7_to_g6	8.474985E-02	0.08475	
Ss_GT_g2_to_g7	6.853773E-07	6.8538E-7	
Ss_GT_g3_to_g7	3.316186E-04	3.3162E-4	
Ss_GT_g4_to_g7	7.979038E-03	0.007979	
Ss_GT_g5_to_g7	4.002267E-02	0.040023	
Ss_GT_g6_to_g7	3.650485E-01	0.36505	

2 Component 1

Date	Jun 28, 2026, 11:01:54 AM
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SETTINGS

Description	Value
Unit system	Same as global system (SI)
Geometry shape function	Automatic

SPATIAL FRAME COORDINATES

First	Second	Third
x	y	z

MATERIAL FRAME COORDINATES

First	Second	Third
X	Y	Z

GEOMETRY FRAME COORDINATES

First	Second	Third
Xg	Yg	Zg

MESH FRAME COORDINATES

First	Second	Third
Xm	Ym	Zm

2.1 DEFINITIONS

2.1.1 Selections

UO2

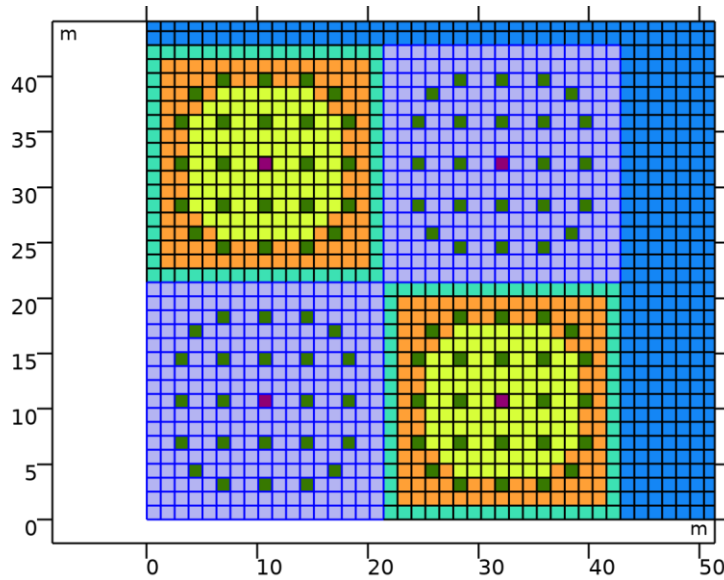
Selection type
Explicit

Selection

Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717

COLOR

Description	Value
Color	Color 12



UO2

MOX %4.3

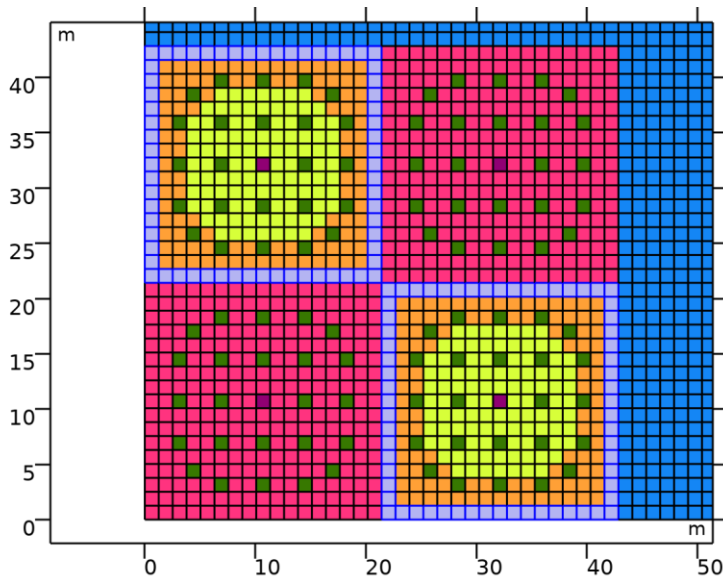
Selection type
Explicit

Selection

Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700

COLOR

Description	Value
Color	Color 10



MOX %4.3

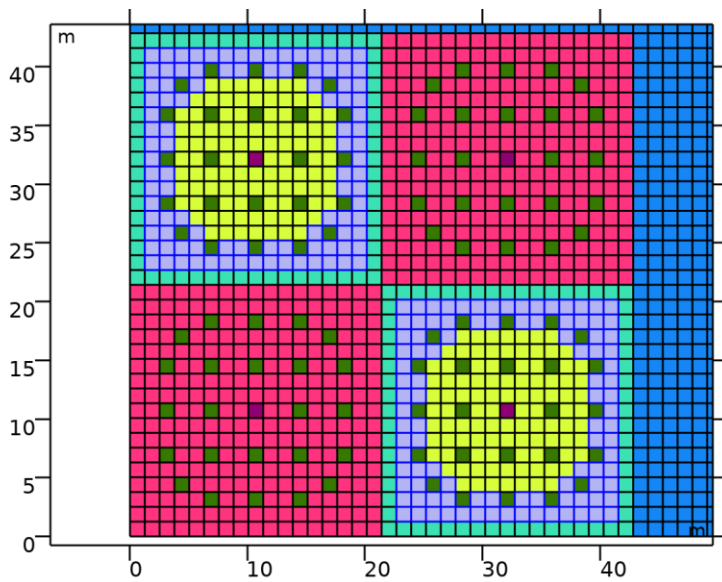
MOX %7.0

Selection type
Explicit

Selection
Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648

COLOR

Description	Value
Color	Color 18



MOX %7.0

MOX %8.7

Selection type

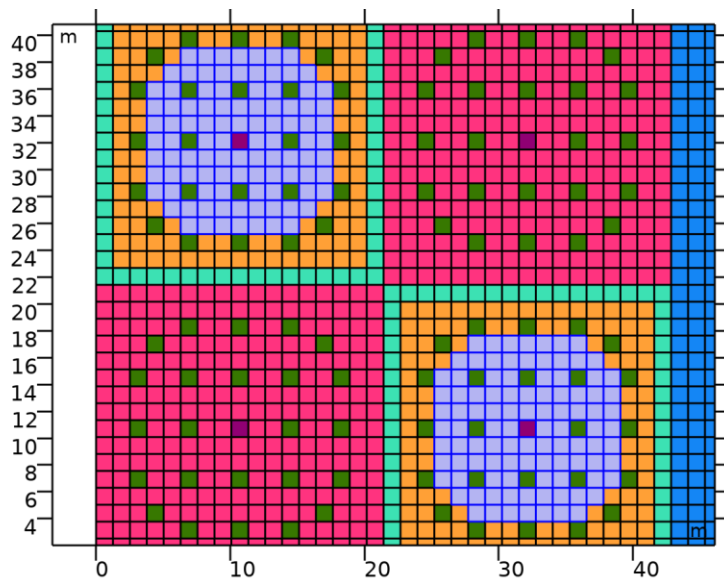
Explicit

Selection

Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542

COLOR

Description	Value
Color	Color 4



MOX %8.7

Fission Chamber

Selection type

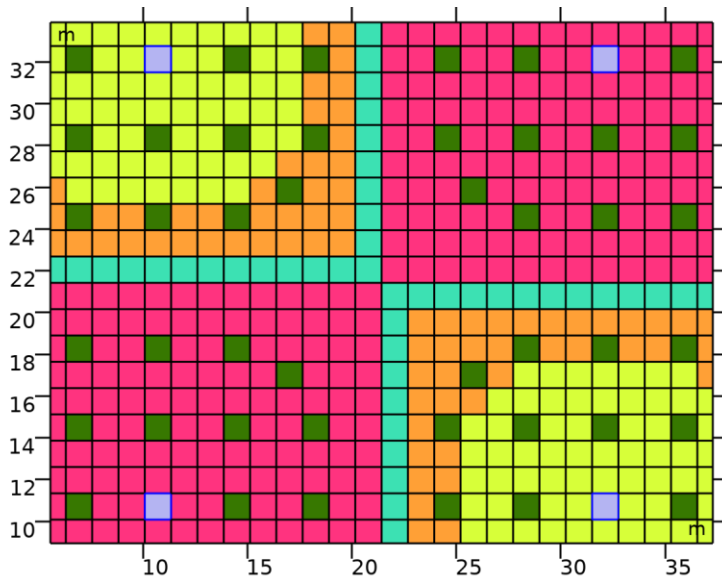
Explicit

Selection

Domains 417, 434, 1284, 1301

COLOR

Description	Value
Color	Color 11



Fission Chamber

Guide Tube

Selection type

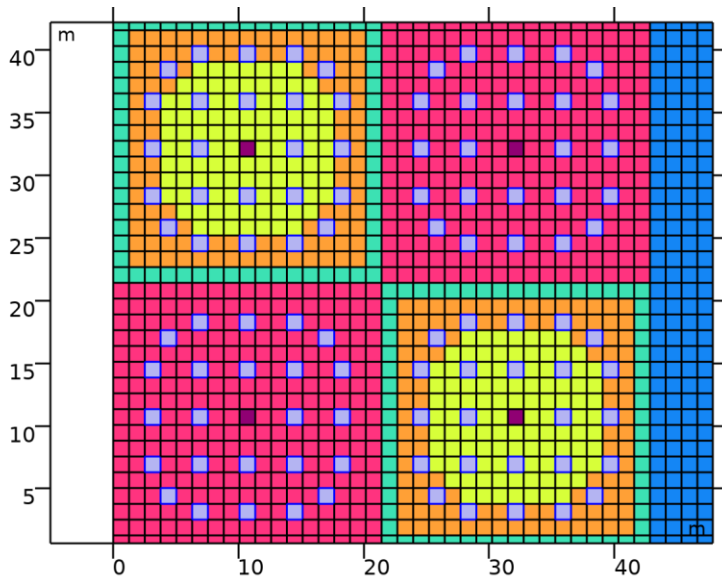
Explicit

Selection

Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610

COLOR

Description	Value
Color	Color 9



Guide Tube

Moderator

Selection type

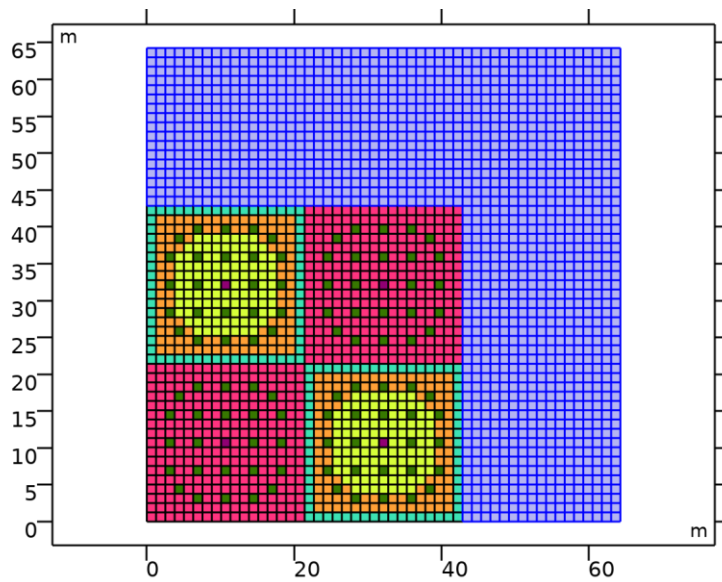
Explicit

Selection

Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601

COLOR

Description	Value
Color	Custom
Custom color	{0.0784313753247261, 0.5254902243614197, 0.9607843160629272}



Moderator

2.1.2 Coordinate Systems

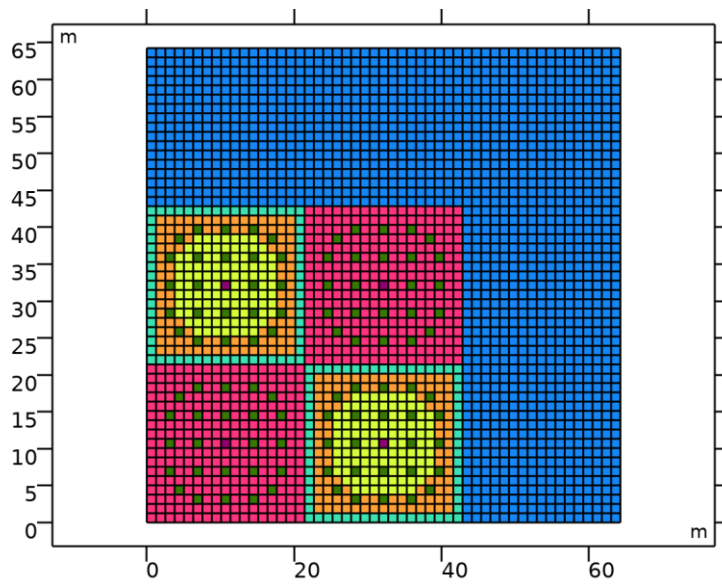
Boundary System 1

Coordinate system type	Boundary system
Tag	sys1

COORDINATE NAMES

First	Second	Third
t1	n	to

2.2 GEOMETRY 1



Geometry 1

UNITS

Length unit	m
Angular unit	deg

GEOMETRY STATISTICS

Description	Value
Space dimension	2
Number of domains	2601
Number of boundaries	5304
Number of vertices	2704

2.2.1 Square 1 (sq1)

POSITION

Description	Value
Position	{0, 0}

SIZE

Description	Value
Side length	1.26

2.2.2 Array 1 (arr1)

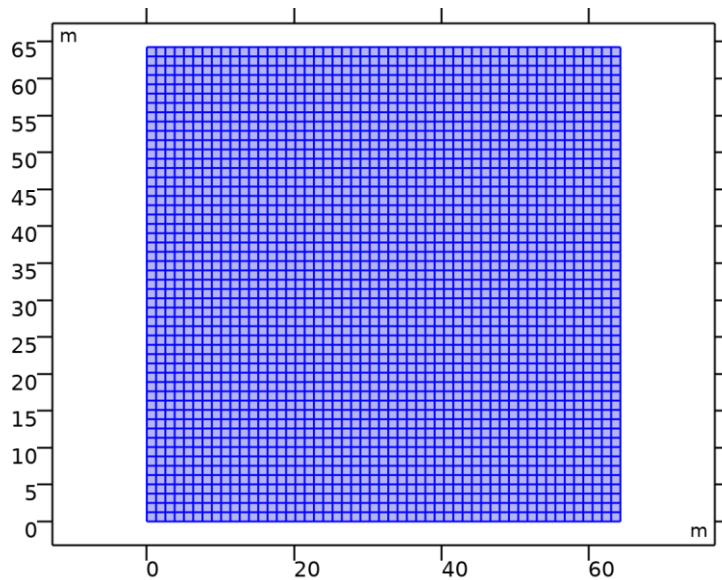
SETTINGS

Description	Value
Size	{51, 51}
Displacement	{1.26, 1.26}

2.3 COEFFICIENT FORM PDE

USED PRODUCTS

COMSOL Multiphysics



Coefficient Form PDE

SELECTION

Geometric entity level	Domain
Selection	Geometry geom1: Dimension 2: All domains

2.3.1 Interface Settings

Discretization

SETTINGS

Description	Value
Shape function type	Lagrange
Element order	Quadratic
Frame	Spatial

SETTINGS

Description	Value
Frequency	From solver

Units

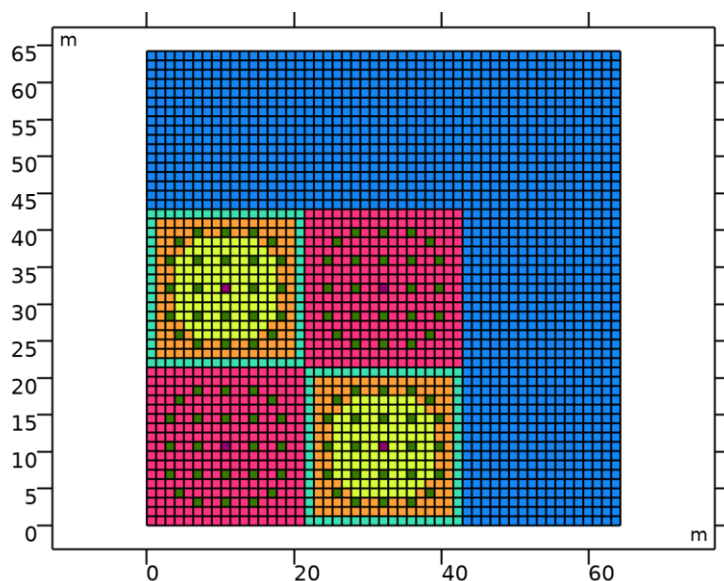
Dependent variable quantity	Unit
Dimensionless	1

Source term quantity	Unit
Custom unit	m ⁻²

2.3.2 Variables

Name	Expression	Unit	Description	Selection	Details
c.nx	nx		Normal vector, x component	Boundaries 1–5304	Meta
c.ny	ny		Normal vector, y component	Boundaries 1–5304	Meta
c.nz	root.nz		Normal vector, z component	Boundaries 1–5304	Meta
c.nxmesh	nxmesh		Normal vector (mesh), x component	Boundaries 1–5304	Meta
c.nymesh	nymesh		Normal vector (mesh), y component	Boundaries 1–5304	Meta
c.nzmesh	root.nzmesh		Normal vector (mesh), z component	Boundaries 1–5304	Meta
c.omega	c.iomega/i	rad/s	Angular frequency	Global	
c.iomega	-lambda	rad/s	Complex angular frequency	Global	
c.freq	0.5*c.omega/pi	Hz	Frequency	Global	

2.3.3 Coefficient Form PDE 1



Coefficient Form PDE 1

SELECTION

Geometric entity level	Domain
Selection	Geometry geom1: Dimension 2: All domains

EQUATIONS

$$\lambda^2 e_a \mathbf{u} - \lambda d_a \mathbf{u} + \nabla \cdot (-c \nabla \mathbf{u} - \alpha \mathbf{u} + \gamma) + \beta \cdot \nabla \mathbf{u} + a \mathbf{u} = f$$

$$\mathbf{u} = [\phi_1, \phi_2, \phi_3, \dots, \phi_7]^T$$

$$\nabla = [\frac{\partial}{\partial x}, \frac{\partial}{\partial y}]$$

SETTINGS

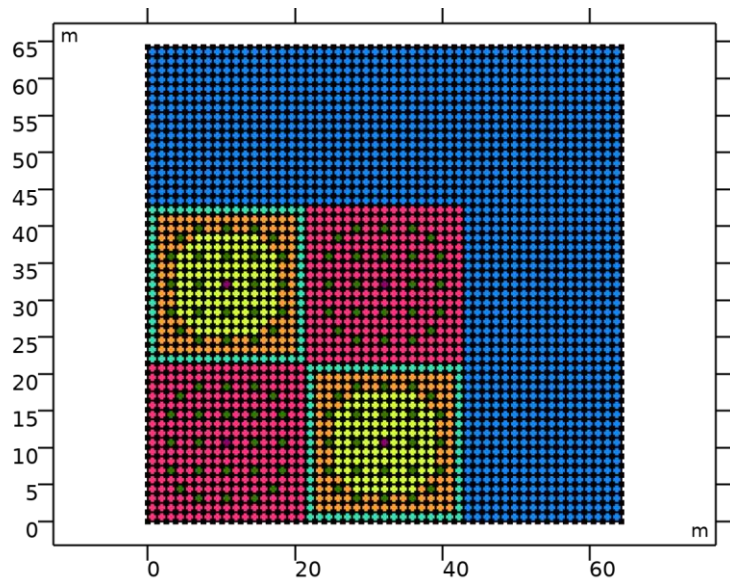
[illegible]

Description	Value
	{0, 0}}, {{0, 0}, {0, 0}}, {{1, 0}, {0, 1}}, {{0, 0}, {0, 0}}, {{{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{1, 0}, {0, 1}}}}
Convection coefficient	{{{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}}}
Source term	{1, 1, 1, 1, 1, 1, 1}
Conservative flux source	{{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}
Conservative flux convection coefficient	{{{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}}}
Mass coefficient	{{0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0}}
Damping or mass coefficient	{{1, 0, 0, 0, 0, 0, 0, 0}, {0, 1, 0, 0, 0, 0, 0, 0}, {0, 0, 1, 0, 0, 0, 0}, {0, 0, 0, 1, 0, 0, 0, 0}, {0, 0, 0, 0, 1, 0, 0, 0}, {0, 0, 0, 0, 0, 1, 0}, {0, 0, 0, 0, 0, 0, 0, 1}}

Shape functions

Name	Shape function	Unit	Description	Shape frame	Selection
phi1	Lagrange (Quadratic)	1	Dependent variable phi1	Spatial	No domains
phi2	Lagrange (Quadratic)	1	Dependent variable phi2	Spatial	No domains
phi3	Lagrange (Quadratic)	1	Dependent variable phi3	Spatial	No domains
phi4	Lagrange (Quadratic)	1	Dependent variable phi4	Spatial	No domains
phi5	Lagrange (Quadratic)	1	Dependent variable phi5	Spatial	No domains
phi6	Lagrange (Quadratic)	1	Dependent variable phi6	Spatial	No domains
phi7	Lagrange (Quadratic)	1	Dependent variable phi7	Spatial	No domains

2.3.4 Zero Flux 1



Zero Flux 1

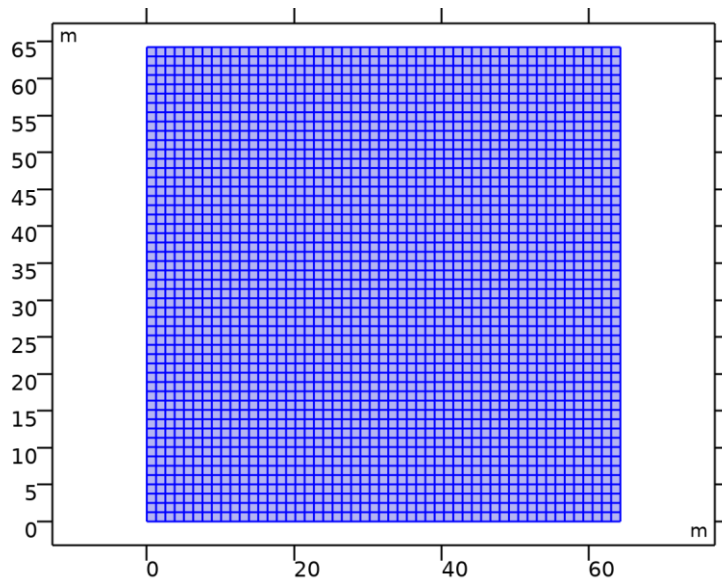
SELECTION

Geometric entity level	Boundary
Selection	Geometry geom1: Dimension 1: All boundaries

EQUATIONS

$$-\mathbf{n} \cdot (-c \nabla \mathbf{u} - \alpha \mathbf{u} + \gamma) = 0$$
$$\mathbf{u} = [\phi i1, \phi i2, \phi i3, \dots, \phi i7]^T$$
$$\nabla = \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right]$$

2.3.5 Initial Values 1



Initial Values 1

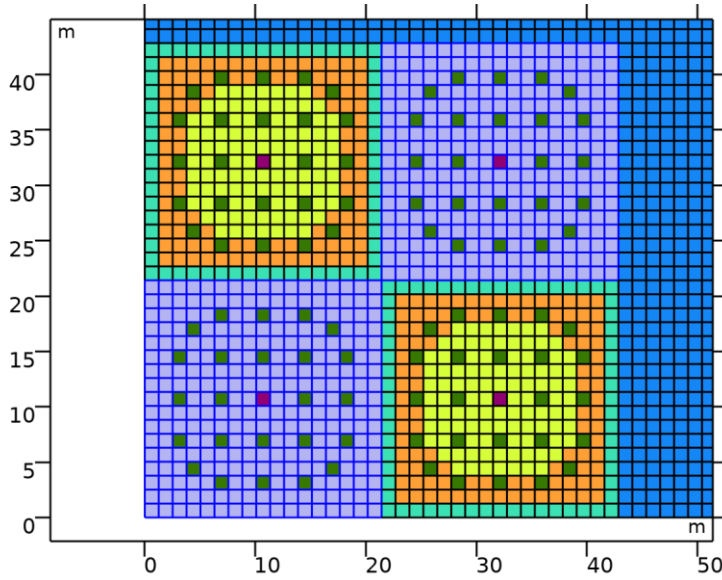
SELECTION

Geometric entity level	Domain
Selection	Geometry geom1: Dimension 2: All domains

SETTINGS

Description	Value
Initial time derivative of phi2	0
Initial time derivative of phi3	0
Initial time derivative of phi4	0
Initial time derivative of phi5	0
Initial time derivative of phi1	0
Initial value for phi5	0
Initial value for phi6	0
Initial value for phi7	0
Initial value for phi1	0
Initial time derivative of phi6	0
Initial value for phi2	0
Initial time derivative of phi7	0
Initial value for phi3	0
Initial value for phi4	0

2.3.6 UO2_PDE



UO2_PDE

SELECTION

Geometric entity level	Domain
Name	UO2
Selection	Named sel1: Geometry geom1: Dimension 2: Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717

EQUATIONS

$$\lambda^2 \mathbf{e}_a \mathbf{u} - \lambda d_a \mathbf{u} + \nabla \cdot (-c \nabla \mathbf{u} - \alpha \mathbf{u} + \gamma) + \beta \cdot \nabla \mathbf{u} + a \mathbf{u} = f$$

$$\mathbf{u} = [\phi i1, \phi i2, \phi i3, \dots, \phi i7]^T$$

$$\nabla = \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right]$$

SETTINGS

Description	Value
-------------	-------

Name	Expression	Unit	Description	Selection
				766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
domflux.phi1y	-D_UO2_g1*phi1y	1/m	Domain flux, y component	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722,

Name	Expression	Unit	Description	Selection
				724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
domflux.phi2x	-D_UO2_g2*phi2x	1/m	Domain flux, x component	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680,

Name	Expression	Unit	Description	Selection
				715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
domflux.phi2y	-D_UO2_g2*phi2y	1/m	Domain flux, y component	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666,

Name	Expression	Unit	Description	Selection
				668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
domflux.phi3x	-D_UO2_g3*phi3x	1/m	Domain flux, x component	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578,

Name	Expression	Unit	Description	Selection
				613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
domflux.phi3y	-D_UO2_g3*phi3y	1/m	Domain flux, y component	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572,

Name	Expression	Unit	Description	Selection
				574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
domflux.phi4x	-D_UO2_g4*phi4x	1/m	Domain flux, x component	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566,

Name	Expression	Unit	Description	Selection
				568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
domflux.phi4y	-D_UO2_g4*phi4y	1/m	Domain flux, y component	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527,

Name	Expression	Unit	Description	Selection
				562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
domflux.phi5x	-D_UO2_g5*phi5x	1/m	Domain flux, x component	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425,

Name	Expression	Unit	Description	Selection
				460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
domflux.phi5y	-D_UO2_g5*phi5y	1/m	Domain flux, y component	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419,

Name	Expression	Unit	Description	Selection
				421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
domflux.phi6x	-D_UO2_g6*phi6x	1/m	Domain flux, x component	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413,

Name	Expression	Unit	Description	Selection
				415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
domflux.phi6y	-D_UO2_g6*phi6y	1/m	Domain flux, y component	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374,

Name	Expression	Unit	Description	Selection
				409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
domflux.phi7x	-D_UO2_g7*phi7x	1/m	Domain flux, x component	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272,

Name	Expression	Unit	Description	Selection
				307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
domflux.phi7y	-D_UO2_g7*phi7y	1/m	Domain flux, y component	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266,

Name	Expression	Unit	Description	Selection
				268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042– 1050, 1052–1054, 1089–1105, 1140– 1141, 1143–1144, 1146–1147, 1149– 1150, 1152–1153, 1155–1156, 1191– 1207, 1242–1258, 1293–1294, 1296– 1297, 1299–1300, 1302–1303, 1305– 1306, 1308–1309, 1344–1360, 1395– 1411, 1446–1447, 1449–1450, 1452– 1453, 1455–1456, 1458–1459, 1461– 1462, 1497–1513, 1548–1550, 1552– 1560, 1562–1564, 1599–1603, 1605– 1606, 1608–1609, 1611–1615, 1650– 1666, 1701–1717

Shape functions

Name	Shape function	Unit	Description	Shape frame	Selection
phi1	Lagrange (Quadratic)	1	Dependent variable phi1	Spatial	Domains 1–17, 52–68, 103–107, 109–110, 112–

Name	Shape function	Unit	Description	Shape frame	Selection
					113, 115–119, 154–156, 158– 166, 168–170, 205–221, 256– 257, 259–260, 262–263, 265– 266, 268–269, 271–272, 307– 323, 358–374, 409–410, 412– 413, 415–416, 418–419, 421– 422, 424–425, 460–476, 511– 527, 562–563, 565–566, 568– 569, 571–572, 574–575, 577– 578, 613–629, 664–666, 668– 676, 678–680, 715–719, 721– 722, 724–725, 727–731, 766– 782, 817–833, 885–901, 936– 952, 987–991, 993–994, 996– 997, 999–1003, 1038–1040, 1042– 1050, 1052–1054, 1089–1105, 1140– 1141, 1143–1144, 1146–1147, 1149– 1150, 1152–1153, 1155–1156, 1191– 1207, 1242–1258, 1293–1294, 1296– 1297, 1299–1300, 1302–1303, 1305– 1306, 1308–1309, 1344–1360, 1395– 1411, 1446–1447, 1449–1450, 1452– 1453, 1455–1456, 1458–1459, 1461– 1462, 1497–1513, 1548–1550, 1552–

Name	Shape function	Unit	Description	Shape frame	Selection
					1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
phi2	Lagrange (Quadratic)	1	Dependent variable phi2	Spatial	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300,

Name	Shape function	Unit	Description	Shape frame	Selection
					1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
phi3	Lagrange (Quadratic)	1	Dependent variable phi3	Spatial	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–

Name	Shape function	Unit	Description	Shape frame	Selection
					1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
phi4	Lagrange (Quadratic)	1	Dependent variable phi4	Spatial	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680,

Name	Shape function	Unit	Description	Shape frame	Selection
					715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
phi5	Lagrange (Quadratic)	1	Dependent variable phi5	Spatial	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170, 205–221, 256–257, 259–260, 262–263, 265–266, 268–269, 271–272, 307–323, 358–374, 409–410, 412–413, 415–416, 418–419, 421–

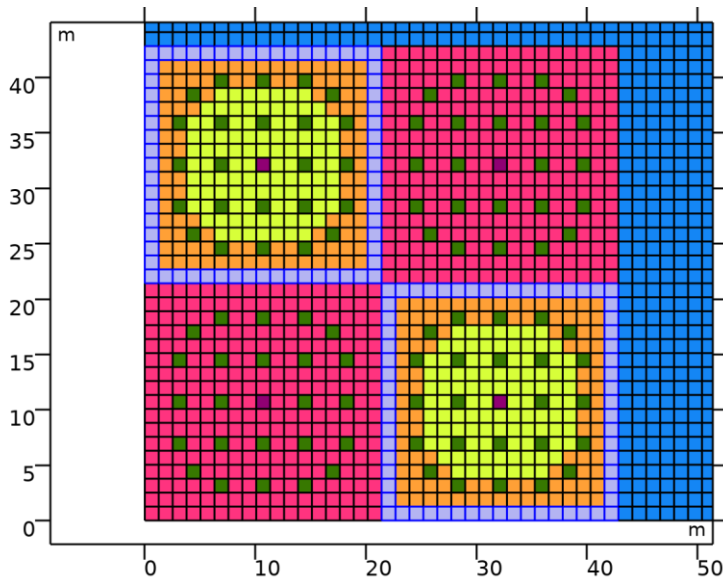
Name	Shape function	Unit	Description	Shape frame	Selection
					422, 424–425, 460–476, 511–527, 562–563, 565–566, 568–569, 571–572, 574–575, 577–578, 613–629, 664–666, 668–676, 678–680, 715–719, 721–722, 724–725, 727–731, 766–782, 817–833, 885–901, 936–952, 987–991, 993–994, 996–997, 999–1003, 1038–1040, 1042–1050, 1052–1054, 1089–1105, 1140–1141, 1143–1144, 1146–1147, 1149–1150, 1152–1153, 1155–1156, 1191–1207, 1242–1258, 1293–1294, 1296–1297, 1299–1300, 1302–1303, 1305–1306, 1308–1309, 1344–1360, 1395–1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717
phi6	Lagrange (Quadratic)	1	Dependent variable phi6	Spatial	Domains 1–17, 52–68, 103–107, 109–110, 112–113, 115–119, 154–156, 158–166, 168–170,

Name	Shape function	Unit	Description	Shape frame	Selection
					205–221, 256– 257, 259–260, 262–263, 265– 266, 268–269, 271–272, 307– 323, 358–374, 409–410, 412– 413, 415–416, 418–419, 421– 422, 424–425, 460–476, 511– 527, 562–563, 565–566, 568– 569, 571–572, 574–575, 577– 578, 613–629, 664–666, 668– 676, 678–680, 715–719, 721– 722, 724–725, 727–731, 766– 782, 817–833, 885–901, 936– 952, 987–991, 993–994, 996– 997, 999–1003, 1038–1040, 1042– 1050, 1052–1054, 1089–1105, 1140– 1141, 1143–1144, 1146–1147, 1149– 1150, 1152–1153, 1155–1156, 1191– 1207, 1242–1258, 1293–1294, 1296– 1297, 1299–1300, 1302–1303, 1305– 1306, 1308–1309, 1344–1360, 1395– 1411, 1446–1447, 1449–1450, 1452– 1453, 1455–1456, 1458–1459, 1461– 1462, 1497–1513, 1548–1550, 1552– 1560, 1562–1564, 1599–1603, 1605– 1606, 1608–1609,

Name	Shape function	Unit	Description	Shape frame	Selection
					1611–1615, 1650–1666, 1701–1717
phi7	Lagrange (Quadratic)	1	Dependent variable phi7	Spatial	Domains 1–17, 52–68, 103–107, 109–110, 112– 113, 115–119, 154–156, 158– 166, 168–170, 205–221, 256– 257, 259–260, 262–263, 265– 266, 268–269, 271–272, 307– 323, 358–374, 409–410, 412– 413, 415–416, 418–419, 421– 422, 424–425, 460–476, 511– 527, 562–563, 565–566, 568– 569, 571–572, 574–575, 577– 578, 613–629, 664–666, 668– 676, 678–680, 715–719, 721– 722, 724–725, 727–731, 766– 782, 817–833, 885–901, 936– 952, 987–991, 993–994, 996– 997, 999–1003, 1038–1040, 1042– 1050, 1052–1054, 1089–1105, 1140– 1141, 1143–1144, 1146–1147, 1149– 1150, 1152–1153, 1155–1156, 1191– 1207, 1242–1258, 1293–1294, 1296– 1297, 1299–1300, 1302–1303, 1305– 1306, 1308–1309, 1344–1360, 1395–

Name	Shape function	Unit	Description	Shape frame	Selection
					1411, 1446–1447, 1449–1450, 1452–1453, 1455–1456, 1458–1459, 1461–1462, 1497–1513, 1548–1550, 1552–1560, 1562–1564, 1599–1603, 1605–1606, 1608–1609, 1611–1615, 1650–1666, 1701–1717

2.3.7 MOX %4.3_PDE



MOX %4.3_PDE

SELECTION

Geometric entity level	Domain
Name	MOX %4.3
Selection	Named sel2: Geometry geom1: Dimension 2: Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700

EQUATIONS

$$\lambda^2 \mathbf{e}_a \mathbf{u} - \lambda d_a \mathbf{u} + \nabla \cdot (-c \nabla \mathbf{u} - \alpha \mathbf{u} + \gamma) + \beta \cdot \nabla \mathbf{u} + a \mathbf{u} = f$$

$$\mathbf{u} = [\phi i1, \phi i2, \phi i3, \dots, \phi i7]^T$$

$$\nabla = [\frac{\partial}{\partial x}, \frac{\partial}{\partial y}]$$

SETTINGS

[illegible]

Name	Expression	Unit	Description	Selection
				289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
domflux.phi1y	-D_MOX43_g1*phi1y	1/m	Domain flux, y component	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
domflux.phi2x	-D_MOX43_g2*phi2x	1/m	Domain flux, x component	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970,

Name	Expression	Unit	Description	Selection
				986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
domflux.phi2y	-D_MOX43_g2*phi2y	1/m	Domain flux, y component	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868– 884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
domflux.phi3x	-D_MOX43_g3*phi3x	1/m	Domain flux, x component	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868– 884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496,

Name	Expression	Unit	Description	Selection
				1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
domflux.phi3y	-D_MOX43_g3*phi3y	1/m	Domain flux, y component	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
domflux.phi4x	-D_MOX43_g4*phi4x	1/m	Domain flux, x component	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
domflux.phi4y	-D_MOX43_g4*phi4y	1/m	Domain flux, y component	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273,

Name	Expression	Unit	Description	Selection
				289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
domflux.phi5x	-D_MOX43_g5*phi5x	1/m	Domain flux, x component	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
domflux.phi5y	-D_MOX43_g5*phi5y	1/m	Domain flux, y component	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970,

Name	Expression	Unit	Description	Selection
				986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
domflux.phi6x	-D_MOX43_g6*phi6x	1/m	Domain flux, x component	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
domflux.phi6y	-D_MOX43_g6*phi6y	1/m	Domain flux, y component	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496,

Name	Expression	Unit	Description	Selection
				1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
domflux.phi7x	-D_MOX43_g7*phi7x	1/m	Domain flux, x component	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
domflux.phi7y	-D_MOX43_g7*phi7y	1/m	Domain flux, y component	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700

Shape functions

Name	Shape function	Unit	Description	Shape frame	Selection
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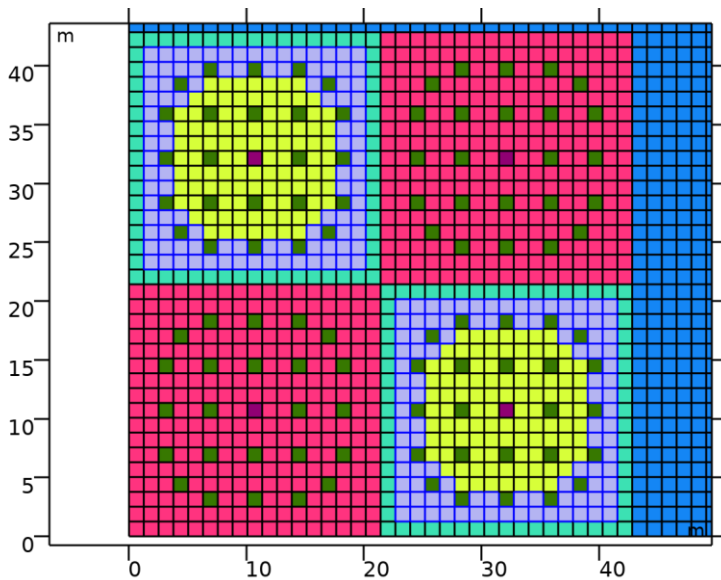
Name	Shape function	Unit	Description	Shape frame	Selection
phi1	Lagrange (Quadratic)	1	Dependent variable phi1	Spatial	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834– 850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
phi2	Lagrange (Quadratic)	1	Dependent variable phi2	Spatial	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834– 850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
phi3	Lagrange	1	Dependent variable	Spatial	Domains 18–34,

Name	Shape function	Unit	Description	Shape frame	Selection
	(Quadratic)		phi3		69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
phi4	Lagrange (Quadratic)	1	Dependent variable phi4	Spatial	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834–850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
phi5	Lagrange (Quadratic)	1	Dependent variable phi5	Spatial	Domains 18–34, 69, 85, 120, 136,

Name	Shape function	Unit	Description	Shape frame	Selection
					171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834– 850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
phi6	Lagrange (Quadratic)	1	Dependent variable phi6	Spatial	Domains 18–34, 69, 85, 120, 136, 171, 187, 222, 238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834– 850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700
phi7	Lagrange (Quadratic)	1	Dependent variable phi7	Spatial	Domains 18–34, 69, 85, 120, 136, 171, 187, 222,

Name	Shape function	Unit	Description	Shape frame	Selection
					238, 273, 289, 324, 340, 375, 391, 426, 442, 477, 493, 528, 544, 579, 595, 630, 646, 681, 697, 732, 748, 783, 799, 834– 850, 868–884, 919, 935, 970, 986, 1021, 1037, 1072, 1088, 1123, 1139, 1174, 1190, 1225, 1241, 1276, 1292, 1327, 1343, 1378, 1394, 1429, 1445, 1480, 1496, 1531, 1547, 1582, 1598, 1633, 1649, 1684–1700

2.3.8 MOX %7.0_PDE



MOX %7.0_PDE

SELECTION

Geometric entity level	Domain
Name	MOX %7.0
Selection	Named sel3: Geometry geom1: Dimension 2: Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–

Description	Value
	Chi_g4*NuSigF_MOX70_g6, 0, 0, 0}, {Chi_g1*NuSigF_MOX70_g7, Chi_g2*NuSigF_MOX70_g7, Chi_g3*NuSigF_MOX70_g7, Chi_g4*NuSigF_MOX70_g7, 0, 0, 0}}

Variables

Name	Expression	Unit	Description	Selection
domflux.phi1x	-D_MOX70_g1*phi1x	1/m	Domain flux, x component	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
domflux.phi1y	-D_MOX70_g1*phi1y	1/m	Domain flux, y component	Domains 70–84, 121–124, 126–127, 129–

Name	Expression	Unit	Description	Selection
				130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
domflux.phi2x	-D_MOX70_g2*phi2x	1/m	Domain flux, x component	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–

Name	Expression	Unit	Description	Selection
				696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
domflux.phi2y	-D_MOX70_g2*phi2y	1/m	Domain flux, y component	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–

Name	Expression	Unit	Description	Selection
				1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
domflux.phi3x	-D_MOX70_g3*phi3x	1/m	Domain flux, x component	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597,

Name	Expression	Unit	Description	Selection
				1634–1648
domflux.phi3y	-D_MOX70_g3*phi3y	1/m	Domain flux, y component	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
domflux.phi4x	-D_MOX70_g4*phi4x	1/m	Domain flux, x component	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–

Name	Expression	Unit	Description	Selection
				543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
domflux.phi4y	-D_MOX70_g4*phi4y	1/m	Domain flux, y component	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075,

Name	Expression	Unit	Description	Selection
				1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
domflux.phi5x	-D_MOX70_g5*phi5x	1/m	Domain flux, x component	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–

Name	Expression	Unit	Description	Selection
				1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
domflux.phi5y	-D_MOX70_g5*phi5y	1/m	Domain flux, y component	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
domflux.phi6x	-D_MOX70_g6*phi6x	1/m	Domain flux, x component	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–

Name	Expression	Unit	Description	Selection
				377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
domflux.phi6y	-D_MOX70_g6*phi6y	1/m	Domain flux, y component	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–

Name	Expression	Unit	Description	Selection
				985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
domflux.phi7x	-D_MOX70_g7*phi7x	1/m	Domain flux, x component	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444,

Name	Expression	Unit	Description	Selection
				1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
domflux.phi7y	-D_MOX70_g7*phi7y	1/m	Domain flux, y component	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648

Shape functions

Name	Shape function	Unit	Description	Shape frame	Selection
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Name	Shape function	Unit	Description	Shape frame	Selection
phi1	Lagrange (Quadratic)	1	Dependent variable phi1	Spatial	Domains 70–84, 121–124, 126– 127, 129–130, 132–135, 172– 173, 175, 183, 185–186, 223– 225, 235–237, 274, 288, 325– 326, 338–339, 376–377, 389– 390, 427, 441, 478–479, 491– 492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682– 683, 685, 693, 695–696, 733– 736, 738–739, 741–742, 744– 747, 784–798, 920–934, 971– 974, 976–977, 979–980, 982– 985, 1022–1023, 1025, 1033, 1035– 1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226– 1227, 1239–1240, 1277, 1291, 1328– 1329, 1341–1342, 1379–1380, 1392– 1393, 1430, 1444, 1481–1483, 1493– 1495, 1532–1533, 1535, 1543, 1545– 1546, 1583–1586, 1588–1589, 1591– 1592, 1594–1597, 1634–1648
phi2	Lagrange (Quadratic)	1	Dependent variable phi2	Spatial	Domains 70–84, 121–124, 126– 127, 129–130, 132–135, 172– 173, 175, 183,

Name	Shape function	Unit	Description	Shape frame	Selection
					185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
phi3	Lagrange (Quadratic)	1	Dependent variable phi3	Spatial	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–

Name	Shape function	Unit	Description	Shape frame	Selection
					390, 427, 441, 478–479, 491– 492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682– 683, 685, 693, 695–696, 733– 736, 738–739, 741–742, 744– 747, 784–798, 920–934, 971– 974, 976–977, 979–980, 982– 985, 1022–1023, 1025, 1033, 1035– 1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226– 1227, 1239–1240, 1277, 1291, 1328– 1329, 1341–1342, 1379–1380, 1392– 1393, 1430, 1444, 1481–1483, 1493– 1495, 1532–1533, 1535, 1543, 1545– 1546, 1583–1586, 1588–1589, 1591– 1592, 1594–1597, 1634–1648
phi4	Lagrange (Quadratic)	1	Dependent variable phi4	Spatial	Domains 70–84, 121–124, 126– 127, 129–130, 132–135, 172– 173, 175, 183, 185–186, 223– 225, 235–237, 274, 288, 325– 326, 338–339, 376–377, 389– 390, 427, 441, 478–479, 491– 492, 529–530, 542–543, 580, 594, 631–633,

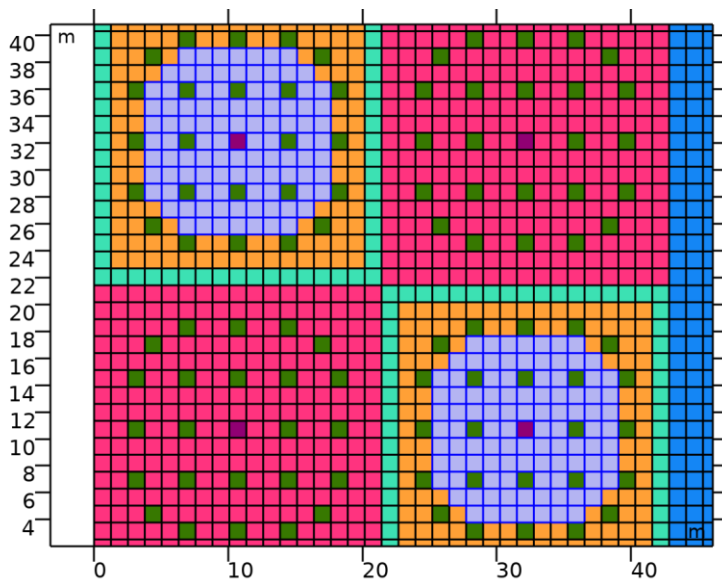
Name	Shape function	Unit	Description	Shape frame	Selection
					643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
phi5	Lagrange (Quadratic)	1	Dependent variable phi5	Spatial	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–

Name	Shape function	Unit	Description	Shape frame	Selection
					747, 784–798, 920–934, 971– 974, 976–977, 979–980, 982– 985, 1022–1023, 1025, 1033, 1035– 1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226– 1227, 1239–1240, 1277, 1291, 1328– 1329, 1341–1342, 1379–1380, 1392– 1393, 1430, 1444, 1481–1483, 1493– 1495, 1532–1533, 1535, 1543, 1545– 1546, 1583–1586, 1588–1589, 1591– 1592, 1594–1597, 1634–1648
phi6	Lagrange (Quadratic)	1	Dependent variable phi6	Spatial	Domains 70–84, 121–124, 126– 127, 129–130, 132–135, 172– 173, 175, 183, 185–186, 223– 225, 235–237, 274, 288, 325– 326, 338–339, 376–377, 389– 390, 427, 441, 478–479, 491– 492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682– 683, 685, 693, 695–696, 733– 736, 738–739, 741–742, 744– 747, 784–798, 920–934, 971– 974, 976–977, 979–980, 982– 985, 1022–1023,

Name	Shape function	Unit	Description	Shape frame	Selection
					1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648
phi7	Lagrange (Quadratic)	1	Dependent variable phi7	Spatial	Domains 70–84, 121–124, 126–127, 129–130, 132–135, 172–173, 175, 183, 185–186, 223–225, 235–237, 274, 288, 325–326, 338–339, 376–377, 389–390, 427, 441, 478–479, 491–492, 529–530, 542–543, 580, 594, 631–633, 643–645, 682–683, 685, 693, 695–696, 733–736, 738–739, 741–742, 744–747, 784–798, 920–934, 971–974, 976–977, 979–980, 982–985, 1022–1023, 1025, 1033, 1035–1036, 1073–1075, 1085–1087, 1124, 1138, 1175–1176, 1188–1189, 1226–

Name	Shape function	Unit	Description	Shape frame	Selection
					1227, 1239–1240, 1277, 1291, 1328–1329, 1341–1342, 1379–1380, 1392–1393, 1430, 1444, 1481–1483, 1493–1495, 1532–1533, 1535, 1543, 1545–1546, 1583–1586, 1588–1589, 1591–1592, 1594–1597, 1634–1648

2.3.9 MOX %8.7_PDE



MOX %8.7_PDE

SELECTION

Geometric entity level	Domain
Name	MOX %8.7
Selection	Named sel4: Geometry geom1: Dimension 2: Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542

EQUATIONS

$$\nabla = [\frac{\partial}{\partial x}, \frac{\partial}{\partial y}]$$

Name	Expression	Unit	Description	Selection
				279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542
domflux.phi1y	-D_MOX87_g1*phi1y	1/m	Domain flux, y component	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542
domflux.phi2x	-D_MOX87_g2*phi2x	1/m	Domain flux, x	Domains 176–182,

Name	Expression	Unit	Description	Selection
			component	226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542
domflux.phi2y	-D_MOX87_g2*phi2y	1/m	Domain flux, y component	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542

Name	Expression	Unit	Description	Selection
domflux.phi3x	-D_MOX87_g3*phi3x	1/m	Domain flux, x component	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542
domflux.phi3y	-D_MOX87_g3*phi3y	1/m	Domain flux, y component	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492,

Name	Expression	Unit	Description	Selection
				1536–1542
domflux.phi4x	-D_MOX87_g4*phi4x	1/m	Domain flux, x component	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542
domflux.phi4y	-D_MOX87_g4*phi4y	1/m	Domain flux, y component	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–

Name	Expression	Unit	Description	Selection
				1442, 1484–1492, 1536–1542
domflux.phi5x	-D_MOX87_g5*phi5x	1/m	Domain flux, x component	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126– 1127, 1129–1130, 1132–1133, 1135– 1136, 1177–1187, 1228–1238, 1279– 1280, 1282–1283, 1285–1286, 1288– 1289, 1330–1340, 1381–1391, 1432– 1433, 1435–1436, 1438–1439, 1441– 1442, 1484–1492, 1536–1542
domflux.phi5y	-D_MOX87_g5*phi5y	1/m	Domain flux, y component	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126– 1127, 1129–1130, 1132–1133, 1135– 1136, 1177–1187, 1228–1238, 1279– 1280, 1282–1283, 1285–1286, 1288– 1289, 1330–1340, 1381–1391, 1432– 1433, 1435–1436,

Name	Expression	Unit	Description	Selection
				1438–1439, 1441–1442, 1484–1492, 1536–1542
domflux.phi6x	-D_MOX87_g6*phi6x	1/m	Domain flux, x component	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542
domflux.phi6y	-D_MOX87_g6*phi6y	1/m	Domain flux, y component	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–

Name	Expression	Unit	Description	Selection
				1433, 1435–1436, 1438–1439, 1441– 1442, 1484–1492, 1536–1542
domflux.phi7x	-D_MOX87_g7*phi7x	1/m	Domain flux, x component	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126– 1127, 1129–1130, 1132–1133, 1135– 1136, 1177–1187, 1228–1238, 1279– 1280, 1282–1283, 1285–1286, 1288– 1289, 1330–1340, 1381–1391, 1432– 1433, 1435–1436, 1438–1439, 1441– 1442, 1484–1492, 1536–1542
domflux.phi7y	-D_MOX87_g7*phi7y	1/m	Domain flux, y component	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126– 1127, 1129–1130, 1132–1133, 1135– 1136, 1177–1187, 1228–1238, 1279– 1280, 1282–1283, 1285–1286, 1288– 1289, 1330–1340,

Name	Expression	Unit	Description	Selection
				1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542

Shape functions

Name	Shape function	Unit	Description	Shape frame	Selection
phi1	Lagrange (Quadratic)	1	Dependent variable phi1	Spatial	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542
phi2	Lagrange (Quadratic)	1	Dependent variable phi2	Spatial	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541,

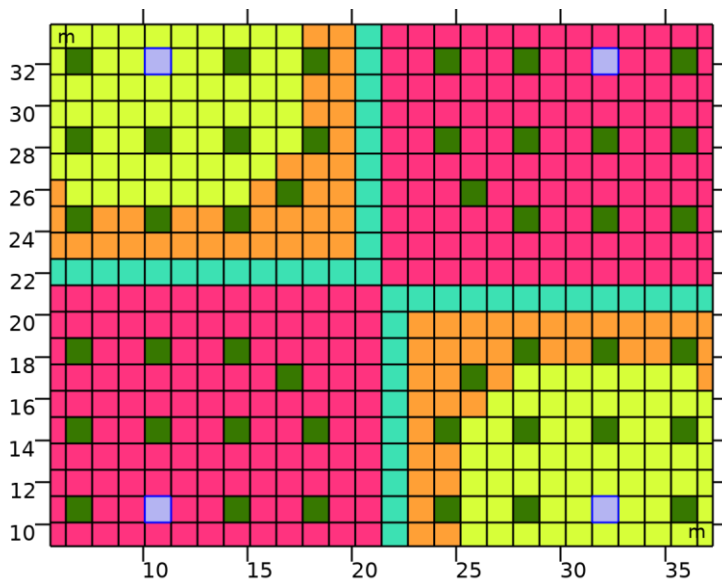
Name	Shape function	Unit	Description	Shape frame	Selection
					582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542
phi3	Lagrange (Quadratic)	1	Dependent variable phi3	Spatial	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542
phi4	Lagrange	1	Dependent variable	Spatial	Domains 176–

Name	Shape function	Unit	Description	Shape frame	Selection
	(Quadratic)		phi4		182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542
phi5	Lagrange (Quadratic)	1	Dependent variable phi5	Spatial	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280,

Name	Shape function	Unit	Description	Shape frame	Selection
					1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542
phi6	Lagrange (Quadratic)	1	Dependent variable phi6	Spatial	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–586, 588–589, 591–592, 634–642, 686–692, 1026–1032, 1076–1084, 1126–1127, 1129–1130, 1132–1133, 1135–1136, 1177–1187, 1228–1238, 1279–1280, 1282–1283, 1285–1286, 1288–1289, 1330–1340, 1381–1391, 1432–1433, 1435–1436, 1438–1439, 1441–1442, 1484–1492, 1536–1542
phi7	Lagrange (Quadratic)	1	Dependent variable phi7	Spatial	Domains 176–182, 226–234, 276–277, 279–280, 282–283, 285–286, 327–337, 378–388, 429–430, 432–433, 435–436, 438–439, 480–490, 531–541, 582–583, 585–

Name	Shape function	Unit	Description	Shape frame	Selection
					586, 588–589, 591–592, 634– 642, 686–692, 1026–1032, 1076– 1084, 1126–1127, 1129–1130, 1132– 1133, 1135–1136, 1177–1187, 1228– 1238, 1279–1280, 1282–1283, 1285– 1286, 1288–1289, 1330–1340, 1381– 1391, 1432–1433, 1435–1436, 1438– 1439, 1441–1442, 1484–1492, 1536– 1542

2.3.10 Fission Chamber_PDE



Fission Chamber_PDE

SELECTION

Geometric entity level	Domain
Name	Fission Chamber
Selection	Named sel5: Geometry geom1: Dimension 2: Domains 417, 434, 1284, 1301

EQUATIONS

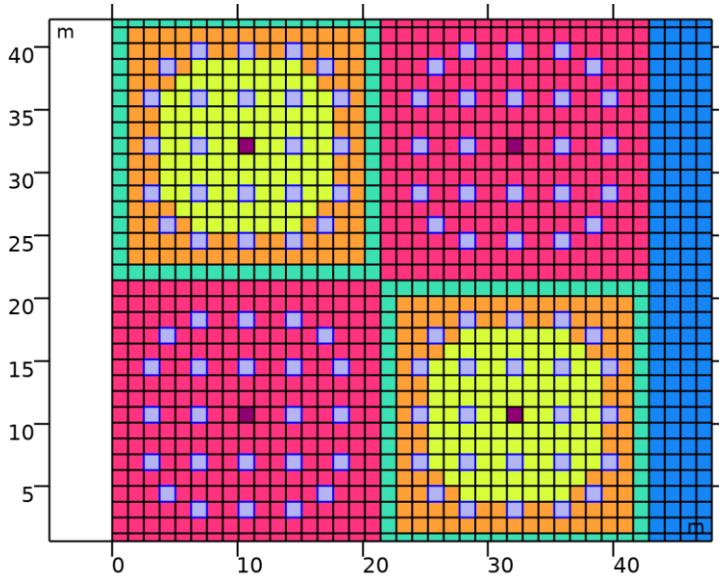
$$\nabla = [\frac{\partial}{\partial x}, \frac{\partial}{\partial y}]$$

Name	Expression	Unit	Description	Selection
			component	1284, 1301
domflux.phi5y	$-D_{FC_g5} \cdot \phi_{5y}$	1/m	Domain flux, y component	Domains 417, 434, 1284, 1301
domflux.phi6x	$-D_{FC_g6} \cdot \phi_{6x}$	1/m	Domain flux, x component	Domains 417, 434, 1284, 1301
domflux.phi6y	$-D_{FC_g6} \cdot \phi_{6y}$	1/m	Domain flux, y component	Domains 417, 434, 1284, 1301
domflux.phi7x	$-D_{FC_g7} \cdot \phi_{7x}$	1/m	Domain flux, x component	Domains 417, 434, 1284, 1301
domflux.phi7y	$-D_{FC_g7} \cdot \phi_{7y}$	1/m	Domain flux, y component	Domains 417, 434, 1284, 1301

Shape functions

Name	Shape function	Unit	Description	Shape frame	Selection
phi1	Lagrange (Quadratic)	1	Dependent variable phi1	Spatial	Domains 417, 434, 1284, 1301
phi2	Lagrange (Quadratic)	1	Dependent variable phi2	Spatial	Domains 417, 434, 1284, 1301
phi3	Lagrange (Quadratic)	1	Dependent variable phi3	Spatial	Domains 417, 434, 1284, 1301
phi4	Lagrange (Quadratic)	1	Dependent variable phi4	Spatial	Domains 417, 434, 1284, 1301
phi5	Lagrange (Quadratic)	1	Dependent variable phi5	Spatial	Domains 417, 434, 1284, 1301
phi6	Lagrange (Quadratic)	1	Dependent variable phi6	Spatial	Domains 417, 434, 1284, 1301
phi7	Lagrange (Quadratic)	1	Dependent variable phi7	Spatial	Domains 417, 434, 1284, 1301

2.3.11 Guide Tube_PDE



Guide Tube_PDE

SELECTION

Geometric entity level	Domain
Name	Guide Tube
Selection	Named sel6: Geometry geom1: Dimension 2: Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610

EQUATIONS

$$\lambda^2 \mathbf{e}_a \mathbf{u} - \lambda d_a \mathbf{u} + \nabla \cdot (-c \nabla \mathbf{u} - \alpha \mathbf{u} + \gamma) + \beta \cdot \nabla \mathbf{u} + a \mathbf{u} = f$$

$$\mathbf{u} = [\phi i1, \phi i2, \phi i3, \dots, \phi i7]^T$$

$$\nabla = \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right]$$

SETTINGS

Description	Value
Absorption coefficient	{{Srem_GT_g1, -Ss_GT_g1_to_g2, -Ss_GT_g1_to_g3, -Ss_GT_g1_to_g4, -Ss_GT_g1_to_g5, 0, 0}, {0, Srem_GT_g2, -Ss_GT_g2_to_g3, -Ss_GT_g2_to_g4, -Ss_GT_g2_to_g5, -Ss_GT_g2_to_g6, -Ss_GT_g2_to_g7}, {0, 0, Srem_GT_g3, -

Description	Value
	Ss_GT_g3_to_g4, -Ss_GT_g3_to_g5, - Ss_GT_g3_to_g6, -Ss_GT_g3_to_g7}, {0, 0, 0, Srem_GT_g4, -Ss_GT_g4_to_g5, - Ss_GT_g4_to_g6, -Ss_GT_g4_to_g7}, {0, 0, 0, - Ss_GT_g5_to_g4, Srem_GT_g5, -Ss_GT_g5_to_g6, -Ss_GT_g5_to_g7}, {0, 0, 0, 0, -Ss_GT_g6_to_g5, Srem_GT_g6, -Ss_GT_g6_to_g7}, {0, 0, 0, 0, 0, - Ss_GT_g7_to_g6, Srem_GT_g7}}
Diffusion coefficient	{{{D_GT_g1, 0}, {0, D_GT_g1}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}}, {{{0, 0}, {0, 0}}, {{D_GT_g2, 0}, {0, D_GT_g2}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{D_GT_g3, 0}, {0, D_GT_g3}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}}, {{{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{D_GT_g4, 0}, {0, D_GT_g4}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{D_GT_g5, 0}, {0, D_GT_g5}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{D_GT_g6, 0}, {0, D_GT_g6}}, {{0, 0}, {0, 0}}, {{{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{0, 0}, {0, 0}}, {{D_GT_g7, 0}, {0, D_GT_g7}}}}
Convection coefficient	{{{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}}
Source term	{0, 0, 0, 0, 0, 0, 0}
Conservative flux source	{{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}
Conservative flux convection coefficient	{{{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}, {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}}}
Mass coefficient	{{0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}}
Damping or mass coefficient	{{0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0,

Description	Value
	0, 0}, {0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0}

Variables

Name	Expression	Unit	Description	Selection
domflux.phi1x	-D_GT_g1*phi1x	1/m	Domain flux, x component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
domflux.phi1y	-D_GT_g1*phi1y	1/m	Domain flux, y component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978,

Name	Expression	Unit	Description	Selection
				981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
domflux.phi2x	-D_GT_g2*phi2x	1/m	Domain flux, x component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
domflux.phi2y	-D_GT_g2*phi2y	1/m	Domain flux, y component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267,

Name	Expression	Unit	Description	Selection
				270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
domflux.phi3x	-D_GT_g3*phi3x	1/m	Domain flux, x component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443,

Name	Expression	Unit	Description	Selection
				1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
domflux.phi3y	-D_GT_g3*phi3y	1/m	Domain flux, y component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
domflux.phi4x	-D_GT_g4*phi4x	1/m	Domain flux, x component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998,

Name	Expression	Unit	Description	Selection
				1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
domflux.phi4y	-D_GT_g4*phi4y	1/m	Domain flux, y component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
domflux.phi5x	-D_GT_g5*phi5x	1/m	Domain flux, x component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281,

Name	Expression	Unit	Description	Selection
				284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
domflux.phi5y	-D_GT_g5*phi5y	1/m	Domain flux, y component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454,

Name	Expression	Unit	Description	Selection
				1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
domflux.phi6x	-D_GT_g6*phi6x	1/m	Domain flux, x component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
domflux.phi6y	-D_GT_g6*phi6y	1/m	Domain flux, y component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041,

Name	Expression	Unit	Description	Selection
				1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
domflux.phi7x	-D_GT_g7*phi7x	1/m	Domain flux, x component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
domflux.phi7y	-D_GT_g7*phi7y	1/m	Domain flux, y component	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414,

Name	Expression	Unit	Description	Selection
				420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610

Shape functions

Name	Shape function	Unit	Description	Shape frame	Selection
phi1	Lagrange (Quadratic)	1	Dependent variable phi1	Spatial	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142,

Name	Shape function	Unit	Description	Shape frame	Selection
					1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
phi2	Lagrange (Quadratic)	1	Dependent variable phi2	Spatial	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
phi3	Lagrange (Quadratic)	1	Dependent variable phi3	Spatial	Domains 108, 111, 114, 125, 128, 131, 157,

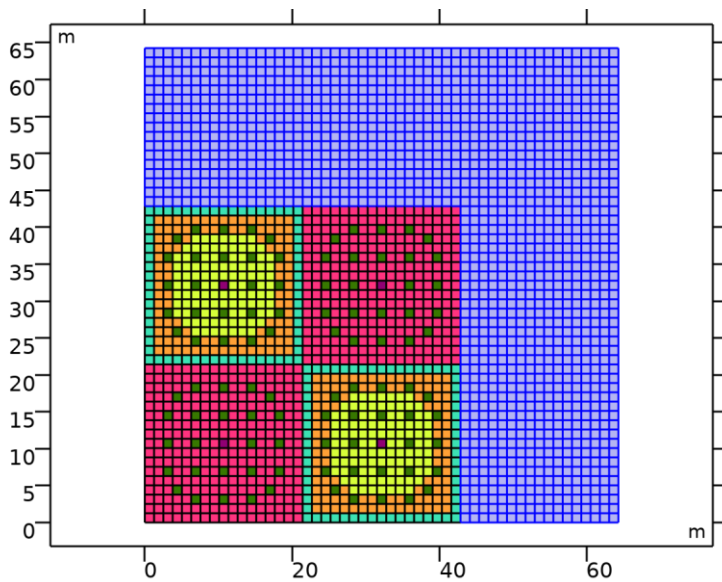
Name	Shape function	Unit	Description	Shape frame	Selection
					167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
phi4	Lagrange (Quadratic)	1	Dependent variable phi4	Spatial	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975,

Name	Shape function	Unit	Description	Shape frame	Selection
					978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
phi5	Lagrange (Quadratic)	1	Dependent variable phi5	Spatial	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587,

Name	Shape function	Unit	Description	Shape frame	Selection
					1590, 1593, 1604, 1607, 1610
phi6	Lagrange (Quadratic)	1	Dependent variable phi6	Spatial	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581, 584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610
phi7	Lagrange (Quadratic)	1	Dependent variable phi7	Spatial	Domains 108, 111, 114, 125, 128, 131, 157, 167, 174, 184, 258, 261, 264, 267, 270, 275, 278, 281, 284, 287, 411, 414, 420, 423, 428, 431, 437, 440, 564, 567, 570, 573, 576, 581,

Name	Shape function	Unit	Description	Shape frame	Selection
					584, 587, 590, 593, 667, 677, 684, 694, 720, 723, 726, 737, 740, 743, 975, 978, 981, 992, 995, 998, 1024, 1034, 1041, 1051, 1125, 1128, 1131, 1134, 1137, 1142, 1145, 1148, 1151, 1154, 1278, 1281, 1287, 1290, 1295, 1298, 1304, 1307, 1431, 1434, 1437, 1440, 1443, 1448, 1451, 1454, 1457, 1460, 1534, 1544, 1551, 1561, 1587, 1590, 1593, 1604, 1607, 1610

2.3.12 Moderator_PDE



Moderator_PDE

SELECTION

Geometric entity level	Domain
Name	Moderator

Name	Expression	Unit	Description	Selection
				1718–2601
domflux.phi1y	-D_Mod_g1*phi1y	1/m	Domain flux, y component	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
domflux.phi2x	-D_Mod_g2*phi2x	1/m	Domain flux, x component	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
domflux.phi2y	-D_Mod_g2*phi2y	1/m	Domain flux, y component	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–

Name	Expression	Unit	Description	Selection
				510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
domflux.phi3x	-D_Mod_g3*phi3x	1/m	Domain flux, x component	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
domflux.phi3y	-D_Mod_g3*phi3y	1/m	Domain flux, y component	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–

Name	Expression	Unit	Description	Selection
				1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
domflux.phi4x	-D_Mod_g4*phi4x	1/m	Domain flux, x component	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
domflux.phi4y	-D_Mod_g4*phi4y	1/m	Domain flux, y component	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683,

Name	Expression	Unit	Description	Selection
				1718–2601
domflux.phi5x	-D_Mod_g5*phi5x	1/m	Domain flux, x component	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
domflux.phi5y	-D_Mod_g5*phi5y	1/m	Domain flux, y component	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
domflux.phi6x	-D_Mod_g6*phi6x	1/m	Domain flux, x component	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–

Name	Expression	Unit	Description	Selection
				510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
domflux.phi6y	-D_Mod_g6*phi6y	1/m	Domain flux, y component	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
domflux.phi7x	-D_Mod_g7*phi7x	1/m	Domain flux, x component	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–

Name	Expression	Unit	Description	Selection
				1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
domflux.phi7y	-D_Mod_g7*phi7y	1/m	Domain flux, y component	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601

Shape functions

Name	Shape function	Unit	Description	Shape frame	Selection
phi1	Lagrange (Quadratic)	1	Dependent variable phi1	Spatial	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–

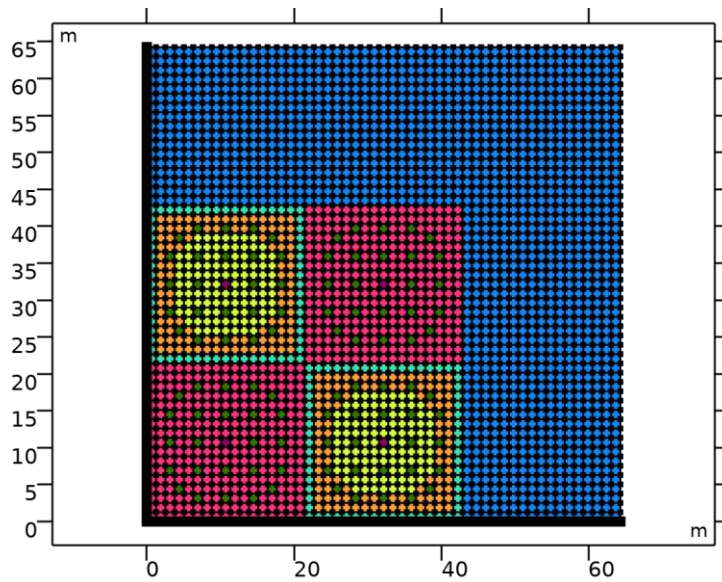
Name	Shape function	Unit	Description	Shape frame	Selection
					1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
phi2	Lagrange (Quadratic)	1	Dependent variable phi2	Spatial	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
phi3	Lagrange (Quadratic)	1	Dependent variable phi3	Spatial	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224,

Name	Shape function	Unit	Description	Shape frame	Selection
					1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
phi4	Lagrange (Quadratic)	1	Dependent variable phi4	Spatial	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
phi5	Lagrange (Quadratic)	1	Dependent variable phi5	Spatial	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–

Name	Shape function	Unit	Description	Shape frame	Selection
					1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
phi6	Lagrange (Quadratic)	1	Dependent variable phi6	Spatial	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377, 1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601
phi7	Lagrange (Quadratic)	1	Dependent variable phi7	Spatial	Domains 35–51, 86–102, 137–153, 188–204, 239–255, 290–306, 341–357, 392–408, 443–459, 494–510, 545–561, 596–612, 647–663, 698–714, 749–765, 800–816, 851–867, 902–918, 953–969, 1004–1020, 1055–1071, 1106–1122, 1157–1173, 1208–1224, 1259–1275, 1310–1326, 1361–1377,

Name	Shape function	Unit	Description	Shape frame	Selection
					1412–1428, 1463–1479, 1514–1530, 1565–1581, 1616–1632, 1667–1683, 1718–2601

2.3.13 Zero Flux 2



Zero Flux 2

SELECTION

Geometric entity level	Boundary
Selection	Geometry geom1: Dimension 1: Boundaries 1–3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 73, 75, 77, 79, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 105, 208, 311, 414, 517, 620, 723, 826, 929, 1032, 1135, 1238, 1341, 1444, 1547, 1650, 1753, 1856, 1959, 2062, 2165, 2268, 2371, 2474, 2577, 2680, 2783, 2886, 2989, 3092, 3195, 3298, 3401, 3504, 3607, 3710, 3813, 3916, 4019, 4122, 4225, 4328, 4431, 4534, 4637, 4740, 4843, 4946, 5049, 5152

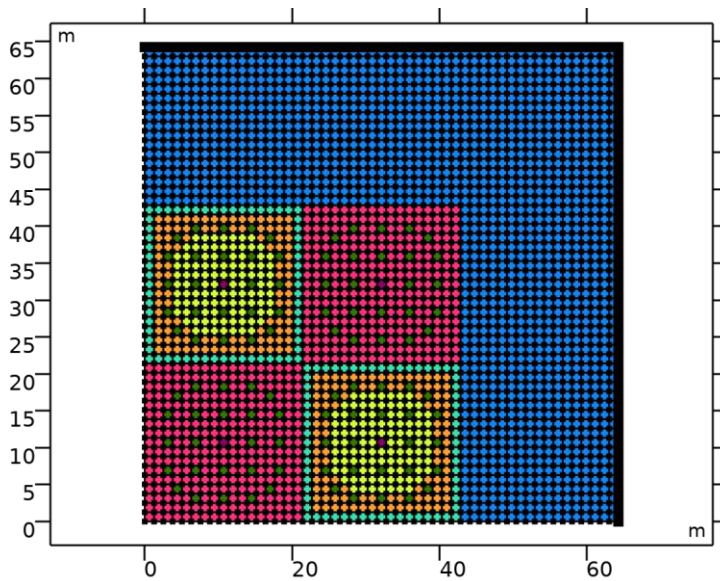
EQUATIONS

$$-\mathbf{n} \cdot (-c \nabla \mathbf{u} - \alpha \mathbf{u} + \gamma) = 0$$

$$\mathbf{u} = [\phi i1, \phi i2, \phi i3, \dots, \phi i7]^T$$

$$\nabla = \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right]$$

2.3.14 Flux/Source 1



Flux/Source 1

SELECTION

Geometric entity level	Boundary
Selection	Geometry geom1: Dimension 1: Boundaries 103, 206, 309, 412, 515, 618, 721, 824, 927, 1030, 1133, 1236, 1339, 1442, 1545, 1648, 1751, 1854, 1957, 2060, 2163, 2266, 2369, 2472, 2575, 2678, 2781, 2884, 2987, 3090, 3193, 3296, 3399, 3502, 3605, 3708, 3811, 3914, 4017, 4120, 4223, 4326, 4429, 4532, 4635, 4738, 4841, 4944, 5047, 5150, 5253–5304

EQUATIONS

$$-\mathbf{n} \cdot (-c \nabla \mathbf{u} - \alpha \mathbf{u} + \gamma) = g - q \mathbf{u}$$

$$\mathbf{u} = [\phi_1, \phi_2, \phi_3, \dots, \phi_7]^T$$

$$\nabla = \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right]$$

SETTINGS

Description	Value
Boundary absorption/impedance term	{{0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0}}
Boundary flux/source	{-0.5*phi1, -0.5*phi2, -0.5*phi3, -0.5*phi4, -0.5*phi5, -0.5*phi6, -0.5*phi7}

Variables

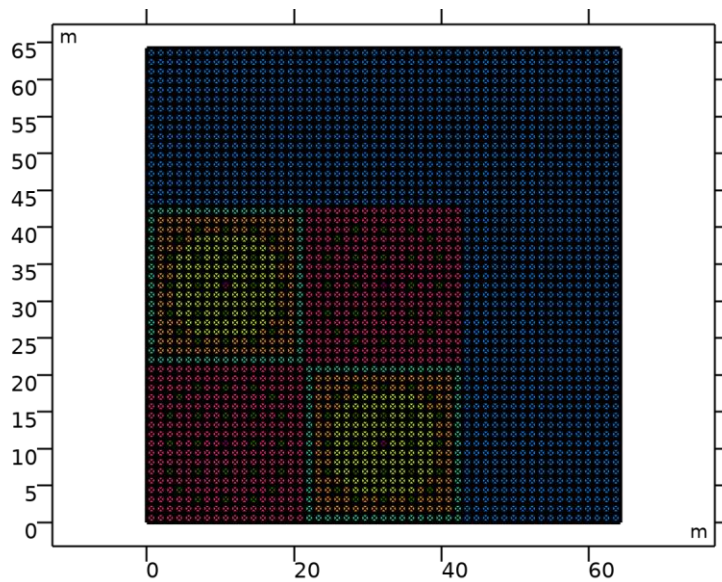
Name	Expression	Unit	Description	Selection	Details
c.g_phi1	-0.5*phi1	1/m	Boundary	Boundaries 103,	+ operation

Name	Expression	Unit	Description	Selection	Details
			flux/source	206, 309, 412, 515, 618, 721, 824, 927, 1030, 1133, 1236, 1339, 1442, 1545, 1648, 1751, 1854, 1957, 2060, 2163, 2266, 2369, 2472, 2575, 2678, 2781, 2884, 2987, 3090, 3193, 3296, 3399, 3502, 3605, 3708, 3811, 3914, 4017, 4120, 4223, 4326, 4429, 4532, 4635, 4738, 4841, 4944, 5047, 5150, 5253–5304	
c.g_phi2	-0.5*phi2	1/m	Boundary flux/source	Boundaries 103, 206, 309, 412, 515, 618, 721, 824, 927, 1030, 1133, 1236, 1339, 1442, 1545, 1648, 1751, 1854, 1957, 2060, 2163, 2266, 2369, 2472, 2575, 2678, 2781, 2884, 2987, 3090, 3193, 3296, 3399, 3502, 3605, 3708, 3811, 3914, 4017, 4120, 4223, 4326, 4429, 4532, 4635, 4738, 4841, 4944, 5047, 5150, 5253–5304	+ operation
c.g_phi3	-0.5*phi3	1/m	Boundary flux/source	Boundaries 103, 206, 309, 412, 515, 618, 721, 824, 927, 1030, 1133, 1236, 1339, 1442, 1545, 1648, 1751, 1854, 1957, 2060, 2163, 2266, 2369, 2472, 2575, 2678, 2781, 2884, 2987, 3090, 3193, 3296, 3399, 3502, 3605, 3708,	+ operation

Name	Expression	Unit	Description	Selection	Details
				3811, 3914, 4017, 4120, 4223, 4326, 4429, 4532, 4635, 4738, 4841, 4944, 5047, 5150, 5253–5304	
c.g_phi4	-0.5*phi4	1/m	Boundary flux/source	Boundaries 103, 206, 309, 412, 515, 618, 721, 824, 927, 1030, 1133, 1236, 1339, 1442, 1545, 1648, 1751, 1854, 1957, 2060, 2163, 2266, 2369, 2472, 2575, 2678, 2781, 2884, 2987, 3090, 3193, 3296, 3399, 3502, 3605, 3708, 3811, 3914, 4017, 4120, 4223, 4326, 4429, 4532, 4635, 4738, 4841, 4944, 5047, 5150, 5253–5304	+ operation
c.g_phi5	-0.5*phi5	1/m	Boundary flux/source	Boundaries 103, 206, 309, 412, 515, 618, 721, 824, 927, 1030, 1133, 1236, 1339, 1442, 1545, 1648, 1751, 1854, 1957, 2060, 2163, 2266, 2369, 2472, 2575, 2678, 2781, 2884, 2987, 3090, 3193, 3296, 3399, 3502, 3605, 3708, 3811, 3914, 4017, 4120, 4223, 4326, 4429, 4532, 4635, 4738, 4841, 4944, 5047, 5150, 5253–5304	+ operation
c.g_phi6	-0.5*phi6	1/m	Boundary flux/source	Boundaries 103, 206, 309, 412, 515, 618, 721, 824, 927, 1030, 1133, 1236,	+ operation

Name	Expression	Unit	Description	Selection	Details
				1339, 1442, 1545, 1648, 1751, 1854, 1957, 2060, 2163, 2266, 2369, 2472, 2575, 2678, 2781, 2884, 2987, 3090, 3193, 3296, 3399, 3502, 3605, 3708, 3811, 3914, 4017, 4120, 4223, 4326, 4429, 4532, 4635, 4738, 4841, 4944, 5047, 5150, 5253– 5304	
c.g_phi7	-0.5*phi7	1/m	Boundary flux/source	Boundaries 103, 206, 309, 412, 515, 618, 721, 824, 927, 1030, 1133, 1236, 1339, 1442, 1545, 1648, 1751, 1854, 1957, 2060, 2163, 2266, 2369, 2472, 2575, 2678, 2781, 2884, 2987, 3090, 3193, 3296, 3399, 3502, 3605, 3708, 3811, 3914, 4017, 4120, 4223, 4326, 4429, 4532, 4635, 4738, 4841, 4944, 5047, 5150, 5253– 5304	+ operation

2.4 MESH 1



Mesh 1

MESH STATISTICS

Description	Value
Status	Complete mesh
Mesh vertices	5305
Triangles	10404
Edge elements	5304
Vertex elements	2704
Number of elements	10404
Minimum element quality	0.75
Average element quality	0.75
Element area ratio	1
Mesh area	4129 m ²

2.4.1 Size (size)

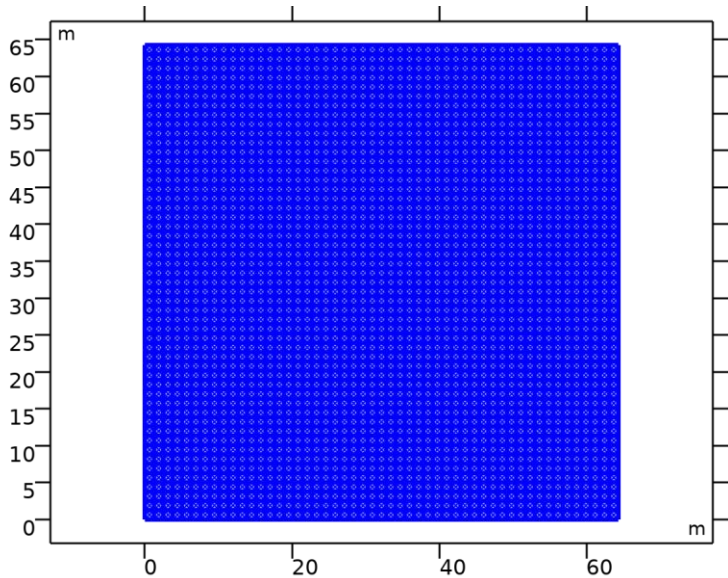
SETTINGS

Description	Value
Maximum element size	4.31
Minimum element size	0.0193
Curvature factor	0.3
Maximum element growth rate	1.3

2.4.2 Free Triangular 1 (ftri1)

SELECTION

Geometric entity level	Domain
Selection	Remaining



Free Triangular 1

SETTINGS

Description	Value
Number of iterations	4
Maximum element depth to process	4
Last build time	4
Built with	COMSOL 6.0.0.318 (win64) 2026 - 06 - 28T14:53:05.253178700

3 Study 1

COMPUTATION INFORMATION

Computation time	47 s
------------------	------

3.1 EIGENVALUE

STUDY SETTINGS

Description	Value
Include geometric nonlinearity	Off

STUDY SETTINGS

Description	Value
Search for eigenvalues around	1
Search for eigenvalues around	On

PHYSICS AND VARIABLES SELECTION

Physics interface	Discretization
Coefficient Form PDE (c)	physics

MESH SELECTION

Geometry	Mesh
Geometry 1 (geom1)	mesh1

3.2 SOLVER CONFIGURATIONS

3.2.1 Solution 1

Compile Equations: Eigenvalue (st1)

STUDY AND STEP

Description	Value
Use study	Study 1
Use study step	Eigenvalue

LOG

```

<---- Compile Equations: Eigenvalue in Study 1/Solution 1 (sol1) -----
Started at Jun 29, 2026, 8:31:53 PM.
Geometry shape function: Quadratic Lagrange
Running on Intel64 Family 6 Model 58 Stepping 9, GenuineIntel.
Using 1 socket with 2 cores in total on DESKTOP-9Q7ES7A.
Available memory: 8.03 GB.
Time: 5 s.
Physical memory: 1.18 GB
Virtual memory: 1.34 GB
Ended at Jun 29, 2026, 8:31:58 PM.
----- Compile Equations: Eigenvalue in Study 1/Solution 1 (sol1) ----->

```

Dependent Variables 1 (v1)

GENERAL

Description	Value
Defined by study step	Eigenvalue

LOG

```

<---- Dependent Variables 1 in Study 1/Solution 1 (sol1) -----
Started at Jun 29, 2026, 8:31:58 PM.
Solution time: 0 s.
Physical memory: 1.19 GB
Virtual memory: 1.35 GB
Ended at Jun 29, 2026, 8:31:58 PM.
----- Dependent Variables 1 in Study 1/Solution 1 (sol1) ----->

```

Dependent variable phi1 (comp1.phi1) (comp1_phi1)

GENERAL

Description	Value
Field components	comp1.phi1
Internal variables	{comp1.uflux.phi1, comp1.dflux.phi1}

Dependent variable phi2 (comp1.phi2) (comp1_phi2)

GENERAL

Description	Value
Field components	comp1.phi2
Internal variables	{comp1.uflux.phi2, comp1.dflux.phi2}

Dependent variable phi3 (comp1.phi3) (comp1_phi3)

GENERAL

Description	Value
Field components	comp1.phi3
Internal variables	{comp1.uflux.phi3, comp1.dflux.phi3}

Dependent variable phi4 (comp1.phi4) (comp1_phi4)

GENERAL

Description	Value
Field components	comp1.phi4
Internal variables	{comp1.uflux.phi4, comp1.dflux.phi4}

Dependent variable phi5 (comp1.phi5) (comp1_phi5)

GENERAL

Description	Value
Field components	comp1.phi5
Internal variables	{comp1.uflux.phi5, comp1.dflux.phi5}

Dependent variable phi6 (comp1.phi6) (comp1_phi6)

GENERAL

Description	Value
Field components	comp1.phi6
Internal variables	{comp1.uflux.phi6, comp1.dflux.phi6}

Dependent variable phi7 (comp1.phi7) (comp1_phi7)

GENERAL

Description	Value
Field components	comp1.phi7
Internal variables	{comp1.uflux.phi7, comp1.dflux.phi7}

Eigenvalue Solver 1 (e1)

GENERAL

Description	Value
Defined by study step	Eigenvalue
Unit	rad/s
Search for eigenvalues around	1

LOG

```

<---- Eigenvalue Solver 1 in Study 1/Solution 1 (sol1) -----
Started at Jun 29, 2026, 8:31:59 PM.
Eigenvalue solver
Number of degrees of freedom solved for: 147091 (plus 218484 internal DOFs).
Nonsymmetric matrix found.
Scales for dependent variables:
Dependent variable phi1 (compl.phi1): 0.008
Dependent variable phi2 (compl.phi2): 0.021
Dependent variable phi3 (compl.phi3): 0.069
Dependent variable phi4 (compl.phi4): 1
Dependent variable phi5 (compl.phi5): 0.06
Dependent variable phi6 (compl.phi6): 0.01
Dependent variable phi7 (compl.phi7): 0.015
Orthonormal null-space function used.
Nonsymmetric eigenvalue solver
Iter      ErrEst  Nconv
   1      2.9e-07   6
20 linear system solutions.
20 matrix multiplications.
19 re-orthogonalizations.
Solution time: 41 s.
Physical memory: 1.95 GB
Virtual memory: 2.19 GB
Ended at Jun 29, 2026, 8:32:40 PM.
----- Eigenvalue Solver 1 in Study 1/Solution 1 (sol1) ----->

```

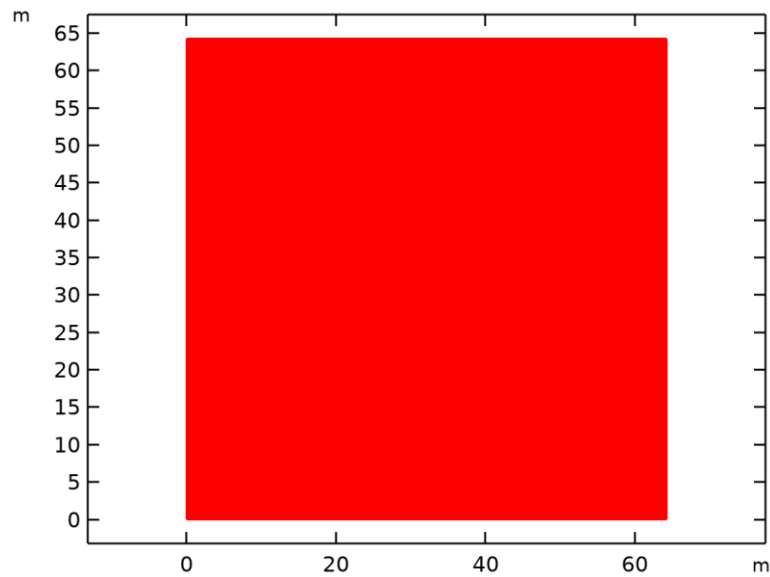
4 Results

4.1 DATASETS

4.1.1 Study 1/Solution 1

SOLUTION

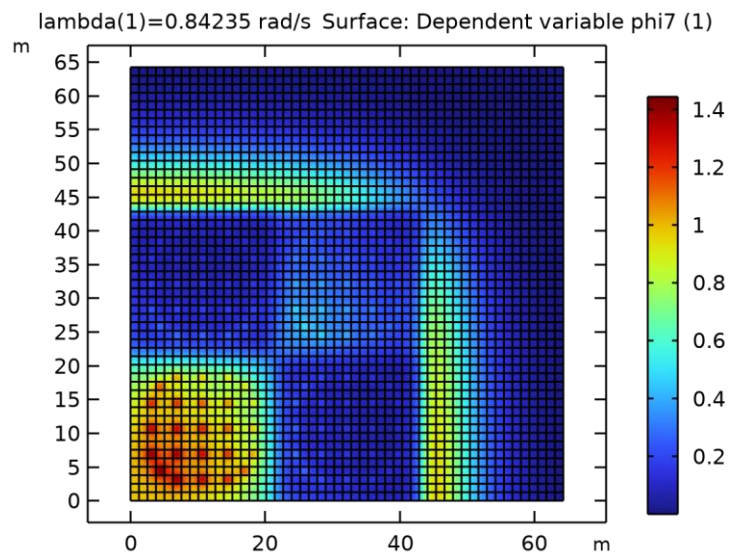
Description	Value
Solution	Solution 1
Component	Component 1 (comp1)



Dataset: Study 1/Solution 1

4.2 PLOT GROUPS

4.2.1 2D Plot Group 1



Surface: Dependent variable ϕ_7 (1)